

Electrical Engineering Material

Chapter 1

THEORY OF METAL

Elementary Quantum Mechanical Ideas

Quantum mechanics is a generalized form of mechanics applicable to very small objects like electron, neutrons etc. in the atom. Classical mechanics are mainly applicable to large objects.

Various phenomena and process that take place within a material cannot be explained without having a good knowledge on quantum mechanics. Although classical mechanics gives ideas about various behavior of different objects, it fails to explain the material behavior for microscopic objects.

Q. Do sub-atomic particle obey Newton's laws of motion?

A. The behavior of the sub-atomic particle cannot be described by Newton's laws.

Wave Particle Duality: De Broglie's Equation

When we deal about sub-atomic particles like electron, photon, neutron we experience that these sub-atomic particle do not follow Newton's laws of motion as well as they do not obey the Maxwell's Electromagnetic Theory as wave. This is because these sub-atomic particles behave as both wave as well as particle.

It was observed that electromagnetic waves eg. X-rays, gamma rays, visible light etc. exhibit as similar to properties of discrete particles of matter.

In 1924, Louis de Broglie presented his research thesis, in which he proposed electrons have properties of both waves and particles, like light. The de Broglie equation is an equation used to describe the wave properties of matter, specifically, the wave nature of the electron:

$$\lambda = h/p$$
$$\lambda = \frac{h}{mv}$$

where λ is wavelength, p is momentum of a particle, h is Planck's constant, m is the mass of a particle, moving at a velocity v .

Also, $p = \hbar k$ and $\hbar = h/2\pi$

From above equations

$k = 2\pi/\lambda$ is a wave number.

The value of $h = 6.624 \times 10^{-34}$ Js

The electromagnetic radiation energy possessed by the photons in the form of packets is given by,

$E = hf$ (known as Plank's equation)

Also we know,

$$\hbar = h/2\pi$$

Thus, above equation can be written as,

$$E = \hbar \times 2\pi f$$

$$E = \hbar \omega \quad (\omega \text{ is known as angular frequency})$$

There were two experiments conducted in early 20th century which were unexplained by classical mechanics and so warranted the need for quantum mechanical ideas.

Experiment 1

It was observed that beams of particle such as electrons were diffracted from a crystal in a manner similar to the diffraction of x-rays, which are electromagnetic waves.

Experiment 2

By bombarding the material with photons of different energies, it was found that for each material a characteristic energy is required to remove the electrons from the surface of material. The minimum frequency of this electromagnetic radiation was found to be,

$$f = \frac{\phi}{h}$$

ϕ is called work function of material

These two experiments proved that the sub-atomic particle possessed the properties of both wave as well as particle. Thus, wave-particle duality was developed.

Wave Function : Schrodinger's Equation

Since, there is no clear distinction between wave and particle in quantum mechanics, there should be some clear representation of what we call wave particle duality.

The displacement about its mean position is represented by wave-function ψ as

$$\psi = A e^{\frac{-j}{\hbar}[Et - px]} \quad \text{--- (i)}$$

Where, A is maximum displacement from mean position or amplitude

$\hbar$ is $h/2\pi$

E is energy of a particle

p is momentum of particle

x is displacement from origin

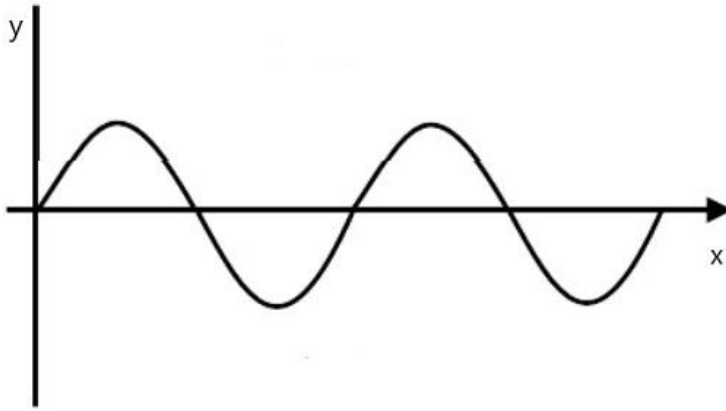


Fig : ordinary wave represented by above equation

Differentiating equation (i) w.r.t x

$$\frac{\partial \psi}{\partial x} = A e^{\frac{-j}{\hbar}[Et - px]} \cdot \frac{jp}{\hbar} \text{ ----- (ii)}$$

Again differentiating equation (ii) w.r.t x

$$\frac{\partial \psi}{\partial x} = A e^{\frac{-j}{\hbar}[Et - px]} \left(\frac{jp}{\hbar} \right)^2$$

$$\frac{\partial \psi}{\partial x} = - \left(\frac{p}{\hbar} \right)^2 \psi \text{ ----- (iii)}$$

For any system total energy E can be written as sum of kinetic energy and potential energy.

$$E = \frac{p^2}{2m} + V$$

$$E - V = \frac{p^2}{2m}$$

$$p^2 = 2m(E - V) \text{ ----- (iv)}$$

From equation (iii) and (iv)

$$\frac{\partial \psi}{\partial x} = - \left(\frac{2m(E - V)}{\hbar^2} \right) \psi$$

$$\frac{\partial \psi}{\partial x} + \left(\frac{2m(E - V)}{\hbar^2} \right) \psi = 0 \text{ ----- (v)}$$

When V is time independent then E is also time independent then equation (v) is known as time independent S.W.E.

In 3D
$$\nabla^2 + \frac{2m[E - v]}{\hbar} \psi = 0$$

Where, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

When V is time dependent then E is also time dependent called SWE.

$$\psi = Ae^{\frac{-j}{\hbar}[Et - px]}$$

Now, differentiating equation (i) wrt t we get,

$$\frac{\partial \psi}{\partial t} = -Ae^{\frac{-j}{\hbar}[Et - px]} \cdot \frac{jE}{\hbar}$$

$$\frac{\partial \psi}{\partial t} = -\frac{jE}{\hbar} \psi$$

$$E\psi = j\hbar \frac{\partial \psi}{\partial t}$$

Now, equation (v) can be written as,

$$\frac{\partial \psi}{\partial x} + \left(\frac{2mE}{\hbar^2}\right) \psi - \left(\frac{2mV}{\hbar^2}\right) \psi = 0$$

$$\frac{\partial \psi}{\partial x} - \left(\frac{2mV}{\hbar^2}\right) \psi = -\left(\frac{2mE}{\hbar^2}\right) E\psi$$

$$\frac{\hbar^2}{2m} \frac{\partial \psi}{\partial x} - V\psi = -j\hbar \frac{\partial \psi}{\partial t}$$

In 3D

$$\frac{\hbar^2}{2m} \nabla^2 \psi - V\psi = -j\hbar \frac{\partial \psi}{\partial t}$$

This is time dependent SWE.

Wave Function (ψ)

The wave can easily be understood and represented in classical mechanics. But the wave function in quantum mechanics does not have similar analogy. The wave function in quantum mechanics used to describe the observable quantities in quantum mechanics.

In quantum mechanics energy, momentum and position of the particle are called observables and the wave function ψ is used to describe these observables. These observables which characterize the particle cannot be measured at the same instant i.e only one can be measured at once.

The wave function is extensively used to predict the probability that a particle would be found in a specified volume of space. Wave function is a complex three dimensional quantity and is dependent upon the time. If $\psi(x,y,z,t)$ represents the wave function then $|\psi|^2$ is called probability amplitude.

The probability of finding the particle in given incremental volume is equal to the probability amplitude (also called probability density function) and incremental volume i.e

$$|\psi(x,y,z)|^2 dx dy dz$$

The probability of locating the particle between x and $x+dx$ in single dimensional case, then the probability will be equal to $|\psi(x)|^2 dx$.

Mathematically, for a particle located somewhere in the universe, we can write

$$\int_{-\infty}^{+\infty} \psi(x,t) dx = 1$$

- Schrodinger's equation does not tell us about the position of the particle rather it only gives us the probability of finding the particle in the vicinity of a point.
- A free particle is that which is not confined to any location. If potential energy $V=0$ in Schrodinger's equation for a particle in one-dimensional case is given by,

$$\psi\psi^* = |A|^2$$

Electron in infinite potential well

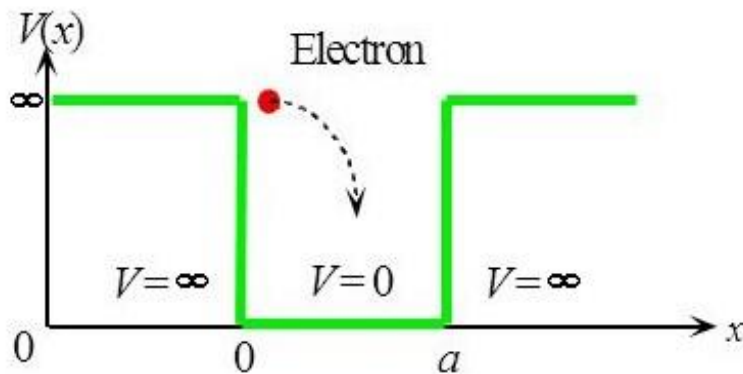


Fig : electron in infinite potential well

Let us consider an electron is present in the region $0 < x < a$ where, a represents the width of the region in which electron is confined. Assume the potential energy of the particle is zero inside the confined region and infinite outside it. The electron thus confined cannot leave the region because to leave, it requires infinite potential energy.

Boundary condition for potential energy is given by

$$V = \begin{cases} 0 & 0 < x < a \\ \infty & 0 \geq x, a \leq x \end{cases}$$

ψ is continuous function, so $\psi(0) = \psi(a) = 0$

Schrodinger's equation inside the box is given by,

$$\frac{\partial^2 \psi}{\partial x^2} + \left(\frac{2mE}{\hbar^2}\right) \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + k^2 \psi = 0 \text{ ----- (i)}$$

Where,

$$k^2 = \left(\frac{2mE}{\hbar^2}\right)$$

General solution for equation (i)

$$\psi(x) = A \sin kx + B \cos kx \text{ (ii)}$$

Where, A and B are some constants which we can find from boundary conditions.

We know from boundary;

$$\psi(x) = 0 \text{ for } x = 0$$

here,

$$0 = A \cdot 0 + B \cdot 1$$

$$B = 0$$

Again,

$$\psi(x) = 0 \text{ for } x = L$$

$$0 = A \sin kL$$

Since, $A \neq 0$ then $\sin kL = 0 = \sin n\pi$

When $n = 1, 2, 3, \dots, \infty$

$$kL = n\pi$$

$$k = \frac{n\pi}{L}$$

Now, from equation (ii)

$$\psi(x) = A \sin \frac{n\pi}{L} x$$

To find the value of A above equation can be normalized as;

$$\int_0^L |\psi|^2 dx = 1$$

$$\int_0^L A^2 \sin^2 \frac{n\pi}{L} x dx = 1$$

$$\int_0^L A^2 \left(1 - 2 \cos \frac{n\pi}{L} x\right) dx = 1$$

$$\frac{A^2}{2} x \Big|_0^L = 1$$

$$A = \left(\frac{2}{L}\right)^{\frac{1}{2}}$$

The normalized wave function of the electron in the box is ;

This is called eigen's function and value is used to measure the total energy of the particle inside the box.

$$k^2 = \left(\frac{2mE}{\hbar^2}\right) = \frac{8\pi^2mE}{h^2}$$

$$E = \frac{k^2 \hbar^2}{8\pi^2 m}$$

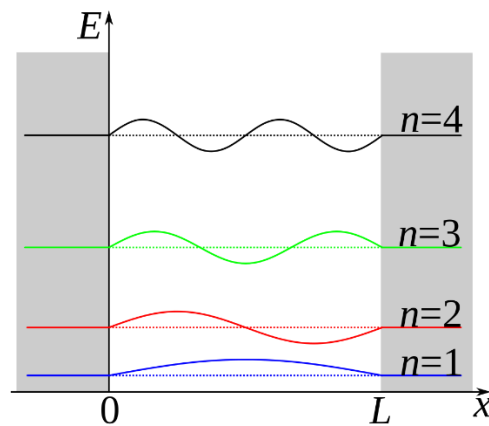


Fig : Energy level and wave function in the well

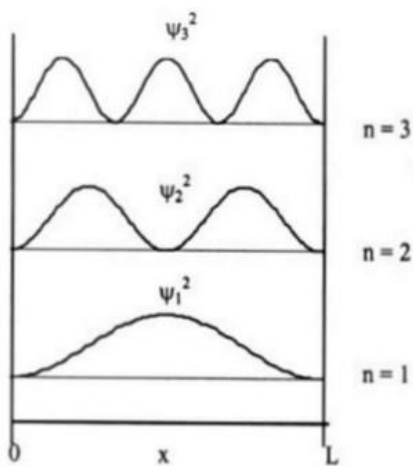


Fig. 3.2. The probability densities in One-dimensional particle-in-a-box

Finite Potential Barrier

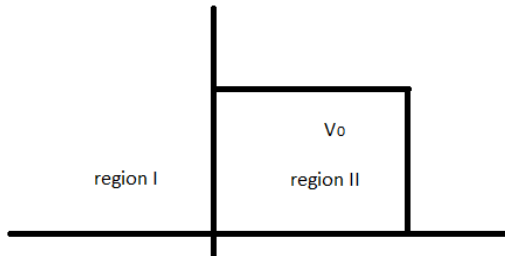


Fig : finite potential barrier

Consider a particle travelling in positive x-direction and encounters a barrier whose potential energy is V_0 . Here, energy of the particle is less than the barrier potential energy also known as barrier height.

When applying Schrodinger's equation for the electron, here, we have to take into account the fact that there are two distinct region where the behavior of electron will be different.

In region 1st electron is free. So, it's potential energy is set equal to zero. So the Schrodinger's equation for region I will be,

$$\frac{\partial \psi_1}{\partial x} + \left(\frac{2mE}{\hbar^2} \right) \psi_1 = 0 \text{ ----- (i) for } x < 0$$

For region II, $x > 0$

We can write Schrodinger's equation as

$$\frac{\partial \psi_2}{\partial x} + \left(\frac{2m(E - V_0)}{\hbar^2} \right) \psi_2 = 0 \text{ -----(ii)}$$

Above equations can be written as,

$$\frac{\partial \psi_1}{\partial x} + \beta^2 \psi_1 = 0 \text{ ----- (iii)}$$

$$\frac{\partial \psi_2}{\partial x} + \alpha^2 \psi_2 = 0 \text{ ----- (iv)}$$

Where,

$$\beta^2 = \frac{2mE}{\hbar^2} \quad \alpha^2 = \frac{2m(E - V_0)}{\hbar^2}$$

The solution of equation (iii) and (iv) can be written as,

$$\psi_1 = Ae^{j\beta x} + Be^{-j\beta x} \text{ -----(v)}$$

$$\psi_2 = Ce^{j\alpha x} + De^{-j\alpha x} \text{ ----- (vi)}$$

Where,

$Ae^{j\beta x}$ = wave travelling in +ve x-direction

$Be^{-j\beta x}$ = wave reflected from potential barrier

$Ce^{-j\alpha x}$ = exponential increasing wave that penetrates the barrier in +ve x-direction

$De^{-j\alpha x}$ = exponential decreasing wave that penetrates the barrier

According to physics, interpretation ψ must be finite only and only if exponential increasing wave does not exists after the penetration of the barrier.

$$\text{i.e. } C=0$$

Now, the constant A, B and D in above equation (v) and (vi) can be found by using boundary condition.

At $x=0$, both the wave functions are continuous. So they must be equal to each other,

$$\psi_1(0) = \psi_2(0)$$

$$Ae^{j\beta 0} + Be^{-j\beta 0} = De^{-j\alpha 0}$$

$$A + B = D \text{ ----- (vii)}$$

The second boundary condition is the slope of both the wave function at $x=0$ are continuous

$$\frac{\partial \psi_1}{\partial x} \bigg|_{x=0} = \frac{\partial \psi_2}{\partial x} \bigg|_{x=0}$$

$$jA\beta e^{j\beta x} - jB\beta e^{-j\beta x} = -jD\alpha e^{-j\alpha x}$$

$$jA\beta e^{j\beta 0} - jB\beta e^{-j\beta 0} = -jD\alpha e^{-j\alpha 0}$$

$$jA\beta - jB\beta = -jD\alpha \text{ ----- (viii)}$$

From (vii) and (viii) we get,

$$A = \frac{D}{2} \left[1 - \frac{\alpha}{\beta} \right]$$

$$B = \frac{D}{2} \left[1 + \frac{\alpha}{\beta} \right]$$

According to the classical mechanics no particle can enter in region IInd but quantum mechanics proves that wave function has a finite value in region IInd wave function decreases exponentially in region IInd.

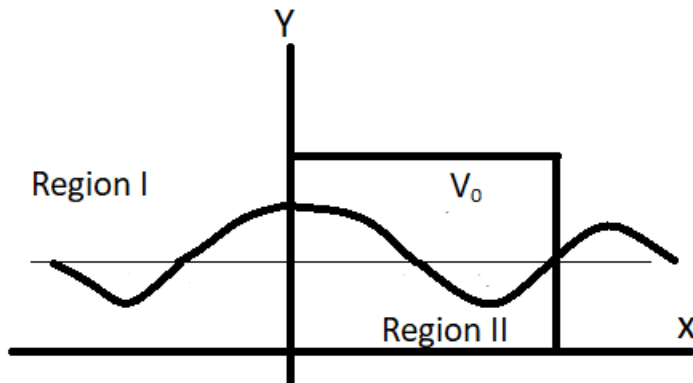


Fig : particle wave function in different region

Tunneling Effect

When beam of particle of energy 'e' is incident on the left of the potential barrier of height ($V_0(V_0 > E)$). According to the classical mechanics the probability of any particle crossing the barrier is impossible. However, according to the quantum mechanics there is probability of the crossing the barrier. This behavior is called tunneling effect.

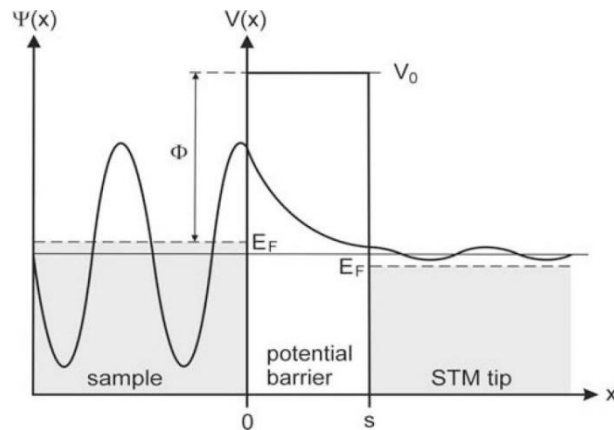


Fig : electron wave length in different region

In region 1, the Schrodinger's equation is

$$\frac{\partial^2 \psi_1}{\partial x^2} + \beta^2 \psi_1 = 0 \text{ ----- (i)}$$

And the solution is,

$$\psi_1 = Ae^{j\beta x} + Be^{-j\beta x} \text{ ----- (ii)}$$

In region 2, the wave function is,

$$\frac{\partial \psi_2}{\partial x} + \alpha^2 \psi_2 = 0 \text{ ----- (iii)}$$

$$\psi_2 = D e^{-j\alpha x} \text{ ----- (iv)}$$

Similarly, in region 3, the Schrodinger's equation is,

$$\frac{\partial \psi_3}{\partial x} + \beta^2 \psi_3 = 0 \text{ ----- (v)}$$

And the solution is,

$$\psi_3 = F e^{j\beta x} + G e^{-j\beta x} \text{ ----- (vi)}$$

Where,

$$\beta^2 = \frac{2mE}{\hbar^2}$$

$$\alpha^2 = \frac{2m(E-V_0)}{\hbar^2}$$

$A e^{j\beta x}$ = wave travelling in +ve x-direction

$B e^{-j\beta x}$ = wave reflected from potential barrier

$D e^{j\alpha x}$ = exponential decrease wave that penetrates the barrier

$F e^{j\beta x}$ = wave travelling in +ve x-direction onwards

$G e^{-j\beta x}$ = wave reflected from region 3 which practically and theoretically does not exist = 0

The relative probability that the electron will turn from region 1 to region 3 through 2 is defined as the transmission coefficient (T) which depends very strongly both on the relative potential barrier height (V_0-E) and the width of the potential barrier. The final expression for the transmission coefficient is expressed as,

$$T = \frac{|\psi_3(x)|^2}{|\psi_1(i)|^2} = \frac{F^2}{A^2}$$

$$T = \frac{1}{1 + D \sinh^2(\alpha L)}$$

$$D = \frac{V_0^2}{4E(V_0-E)}$$

On solving we get,

$$T = \frac{16E(V_0-E)}{V_0^2} e^{-2\alpha L}$$

And the reflection coefficient R is given by,

$$R = \frac{B^2}{A^2} = 1 - T$$

Pauli Exclusion Principle

It states that “no two electrons in any one atom can have all four identical quantum number”.

Where,

n = principle quantum number

l = orbital angular momentum quantum number

m_l = magnetic quantum number

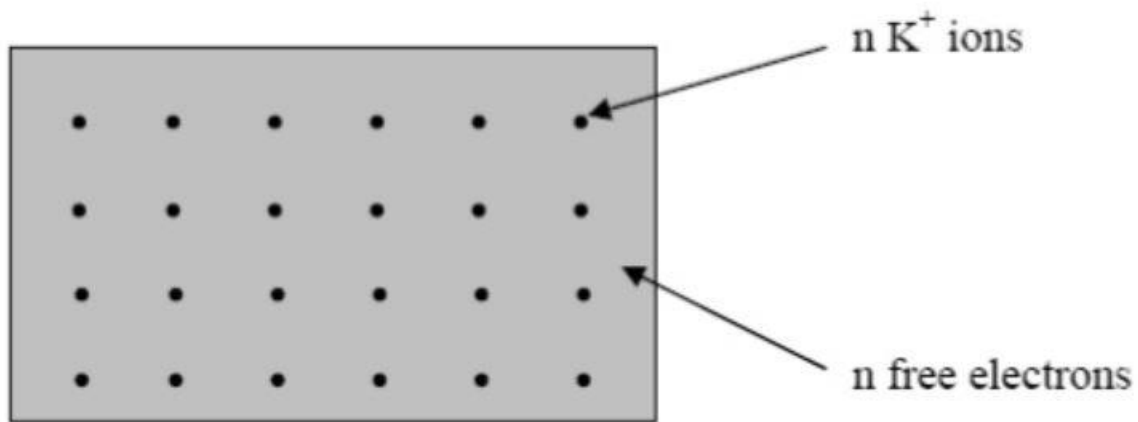
m_s = spin magnetic quantum number

and the wave function is given by, ψ_{n,l,m_l,m_s} .

According to the principle there must be at least one of the quantum number different than other in an electrons of an atom. If four quantum numbers of two electrons are identical, the one of the two electrons will be excluded from the electronic configuration of an atom. Therefore, this principle is called exclusion principle.

Free Electron Theory Of Metal

Drude in 1900 constructed theory of metal on the basis of kinetic theory of gases.



In gas a single type of particle is present but in metal at least two types of particles are present. i.e. electron and protons.

According to this theory the outermost electrons of metal known as valance electrons of atoms are loosely attached to the parent atom and these electron are free to move throughout the whole metal forming gas.

Momentum (p) = mv

Kinetic energy of electrons (KE) = $\frac{1}{2}mv^2 = \frac{p^2}{2m}$

We have,

$$p = \hbar k$$

$$KE = \frac{(\hbar k)^2}{2m}$$

$$KE = ck^2$$

$$\text{Where, } c = \frac{\hbar^2}{2m}$$

Fermi Energy

Fermi energy is defined as the highest filled energy level at a temperature of absolute zero. i.e. 0⁰K. All the energy levels up to the fermi level are filled at 0⁰K and empty above it. The distribution of electron among various energy level in accordance with Pauli's exclusion principle. If N is the total numbers of electrons to be accommodated on the line then for even n, we have $N = 2n_f$. Thus, for $n = n_f$ the expression can be written as,

$$E_F = \frac{\hbar^2}{2m} \left(\frac{\pi n_F}{2} \right)^2 \text{----- (1)}$$

Thus, the fermi energy depends upon the length of the box and the number of electrons in the box.

$$E_F = \frac{\hbar^2}{2m} \left(\frac{\pi N}{2L} \right)^2 = \frac{\hbar^2}{2m} K_F^2$$

Where, K_f = Fermi-wave vector

n_f = principle quantum number of the fermi level

The number of occupied electrons is sphere is given by;

$$N = 2 \times \frac{\text{Vol. of sphere of radius } K_F}{\text{Vol. of element of } K\text{-space}}$$

$$N = 2 \times \frac{\frac{4}{3}\pi K_F^3}{K_x^3}$$

$$N = \frac{L^3 K^3}{3\pi^2} \left(K_x = \frac{2\pi}{L} \right)$$

$$N = \frac{VK^3}{3\pi^2}$$

$$\text{And } n = \frac{N}{V} = \frac{K_f^3}{3\pi^3}$$

Operators

Operators are used to predict the parameters like energy, momentum and position of an electron.

We have, SWE

$$\psi = Ae^{\frac{-j}{\hbar}[Et - px]}$$

Differentiating w.r.t t

$$\frac{\partial \psi}{\partial t} = -Ae^{\frac{-j}{\hbar}[Et - px]} \cdot \frac{jE}{\hbar}$$

$$\frac{\partial \psi}{\partial t} = -\frac{jE}{\hbar} \psi$$

$$E\psi = j\hbar \frac{\partial \psi}{\partial t}$$

This $E \rightarrow j\hbar \frac{\partial}{\partial t}$ is called energy operator.

Similarly, differentiating w.r.t x.

$$\frac{\partial \psi}{\partial x} = Ae^{\frac{-j}{\hbar}[Et - px]} \cdot \frac{jp}{\hbar}$$

Degenerate State

Different combination of principle quantum number representing the same energy is called degenerate state. The energy of electron in potential well is $E = \frac{n^2 \hbar^2}{8mL^2}$. Here, principle quantum number 'n' is taken for single dimension and for three dimension, $n^2 = n_x^2 + n_y^2 + n_z^2$.

Then, the energy can be expressed as, $E_n = \frac{\hbar^2}{8mL^2} n_x^2 + n_y^2 + n_z^2$.

For same value of equation 'n' along x-axis can represent different energy because it also depends on 'n' along y and z-axis.

For example (1,2,1), (2,1,1) and (1,1,2) represent same electron energy. These all three electronic state are degenerate states.

Heisenberg's Uncertainty Principle

It is impossible to determine both position and momentum of an electron simultaneously with perfect accuracy. If an experiment is designate to measure one of them exactly, the other will become uncertain and vice-versa.

Let Δx be the uncertainty to introduce in measuring the position and Δp be the uncertainty to introduce in measuring momentum. If we try to measure both at once then according to uncertainty principle.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Where, $\hbar = \frac{h}{2\pi}$, h Plank's constant.

Definition Of Number Of States, Density Of States

The quantized energy state of electron is ;

$$E = \frac{n^2 \hbar^2}{8mL^2}$$

$$E_n = \frac{\hbar^2}{8mL^2} n_x^2 + n_y^2 + n_z^2$$

Each set of quantum number exists a specific energy level called energy state. For $n_x=n_y=n_z=1$, there is one energy state. So unit cube has one energy state. Hence the number of energy state in any energy volume is proportional to that value.

The principle quantum number takes only positive value so it takes only radius 'n'. The number of energy state within the sphere of radius 'n' is,

$$N = \frac{1}{8} \frac{4\pi n^3}{3}$$

Also,

$$E = \frac{n^2 \hbar^2}{8mL^2}$$

$$8mL^2 E = n^2 \hbar^2$$

$$n = \sqrt{\frac{8mL^2 E}{\hbar^2}}$$

Now,

$$N = \frac{1}{8} \cdot \frac{4\pi}{3} \left[\frac{8mL^2 E}{\hbar^2} \right]^{3/2}$$

$$N = \frac{\pi}{6} \cdot K^{3/2} E^{3/2}$$

Where,

$$k = \frac{8mL^2}{\hbar^2}$$

The number of state represent the number of energy level in quantum space. The density of states is number of energy state per unit energy in an energy interval of dE.

$$\frac{dN}{dE} = \frac{3}{2} \frac{\pi}{6} K^{3/2} E^{1/2}$$

$$\frac{dN}{dE} = \frac{\pi}{4} K^{3/2} E^{1/2}$$

Here, $\frac{dN}{dE} = Z(E) = \text{Density of state}$

When electron is placed in potential box with side 'l' then the volume of box is,

$$V = l^3$$

We have,

$$Z(E) = \frac{dN}{dE} = \frac{\pi}{4} K^{\frac{3}{2}} E^{\frac{1}{2}}$$

$$Z(E) = \frac{dN}{dE} = \frac{\pi}{4} \left[\frac{8mL^2}{h^2} \right]^{\frac{3}{2}} E^{\frac{1}{2}}$$

$$Z(E) = \frac{dN}{dE} = \frac{\pi}{4} 4 \left[\frac{2m}{h^2} \right]^{\frac{3}{2}} E^{\frac{1}{2}} V$$

$$Z(E) = \frac{dN}{dE} = 2\pi V \left[\frac{2m}{h^2} \right]^{\frac{3}{2}} E^{\frac{1}{2}}$$

$$Z(E) = C E^{\frac{1}{2}}$$

$$\text{Where, } C = 2\pi V \left[\frac{2m}{h^2} \right]^{\frac{3}{2}}$$

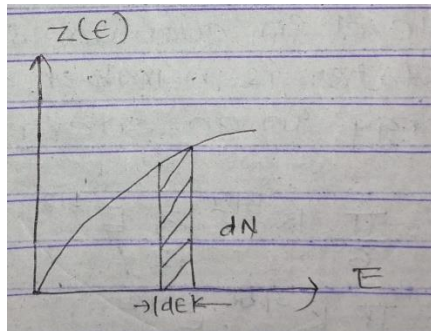


Fig: Density of state vs energy function

In the figure, the curve representing $Z(E)$ is hyperbola where all energy levels less than $E(n)$ are taken into account. The shaded area represents the number of energy state in an elementary energy interval $d(E)$.

$$dN = Z(E) dE$$

Contact Potential

Each metal has different work function which makes their fermi level different. When metal with different fermi level are in contact then electrons from metal with higher fermi level moves towards lower fermi level. This transfer of electron from higher fermi level to lower fermi level makes it

positively charge and metals with low fermi level becomes negatively charge. Finally, a potential difference is developed at junction called contact potential. This process continues till the contact potential is large enough to prevent the further transfer of electron.

This is called system at equilibrium. The contact potential due to difference in work function can be calculated as;

$$e\nabla v = \phi_2 - \phi_1$$

$$\nabla v = \frac{\phi_2 - \phi_1}{e}$$

Thermal Emission And Work Function

The process of ejection of electrons from the surface of the metal at rather high temperature is called thermal emission. There is a threshold(minimum) value of energy which electron must possess if it is to leave the metal surface. The threshold value of energy is called work function of the metal. Work function is denoted by Greek alphabet ϕ .

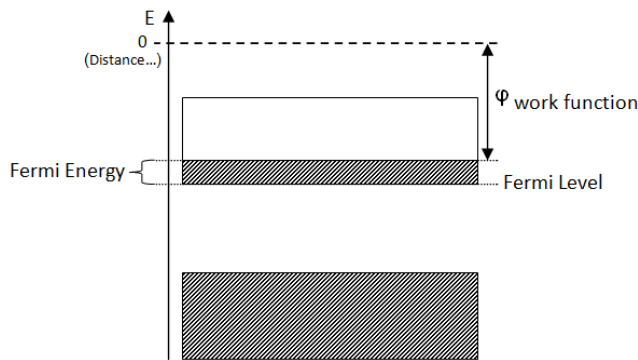


Fig: Fermi-level and work function in metal

In 1901 Richardson published the results of this experiment, the current from a heated wire seemed to depend exponentially on the temperature of the wire. A mathematical form similar to Arrhenius equation. Later he proposed that the emission law should have the mathematical form,

$$I = A_G T^2 e^{-\phi/KT}$$

I is emission current

T is temperature of the metal

ϕ is work function of the metal

K is Boltzman's constant

There was no consensus among interested theoreticians as to what is the exact expression for A_G but there is agreement that A_G must be written in the form,

$$A_G = \lambda_R A_0$$

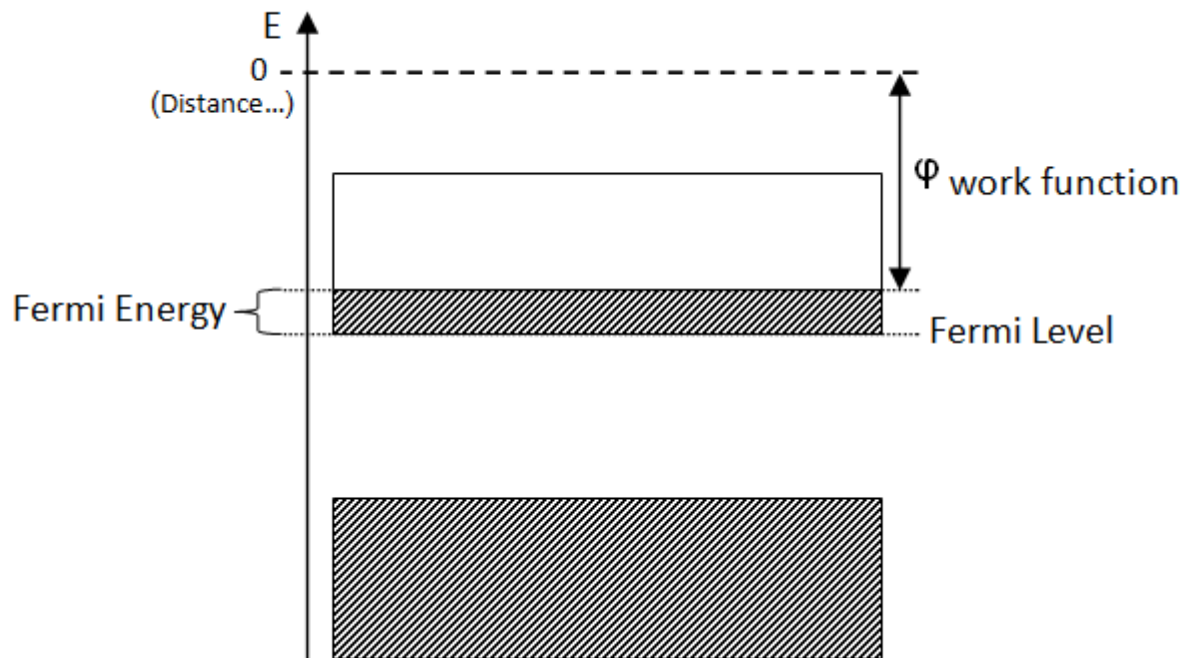
Where, λ_R is a material specific correction factor that is typically of order 0.5 and A_0 is a universal constant given by,

$$A_0 = \frac{4\pi m k^2 e}{h^3}$$

Where, m and e are the mass and charge of an electron and h is plank's constant.

Fermi-Dirac Distribution

Let us consider state of two particle with wave function ψ_1 and ψ_2 and their corresponding energy E_1 and E_2 . This two particles goes on interaction or collision and they change their state.



Now, above particles have wave function ψ_3 and ψ_4 and their corresponding energy value E_3 and E_4 respectively after collision.

Now, after collision,

$$f(E_1)f(E_2)[1 - f(E_3)][1 - f(E_4)] = f(E_3)f(E_4)[1 - f(E_1)][1 - f(E_2)] \text{----- (1)}$$

$$\text{Forward process} = \text{Backward process}$$

$$E_1 + E_2 = E_3 + E_4$$

Now, the fermi-Dirac distribution function is given by,

$$F(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

Where,

$F(E)$ = Occupation index or probability of occupation

E = Energy of the electron

E_F = Fermi energy level

T = Absolute temperature

k = Boltzmann's constant

Case I : When $T=0$ and $E_F \gg E$

$$F(E) = \frac{1}{1+e^{-\infty}}$$

$$F(E) = 1$$

Case II : When $T=0$ and $E_F \ll E$

$$F(E) = \frac{1}{1+e^{\infty}}$$

$$F(E) = 0$$

Case III : When $T \neq 0$, $E_F = E$

$$F(E) = \frac{1}{1+e^0} = \frac{1}{2}$$

Case IV : When $T \neq 0$, $E_F < E$

$$F(E) = \frac{1}{1+e^{E/kT}} = \frac{1}{e^{E/kT}}$$

By using Fermi function or occupation index we can calculate the number of energy occupied by the electron.

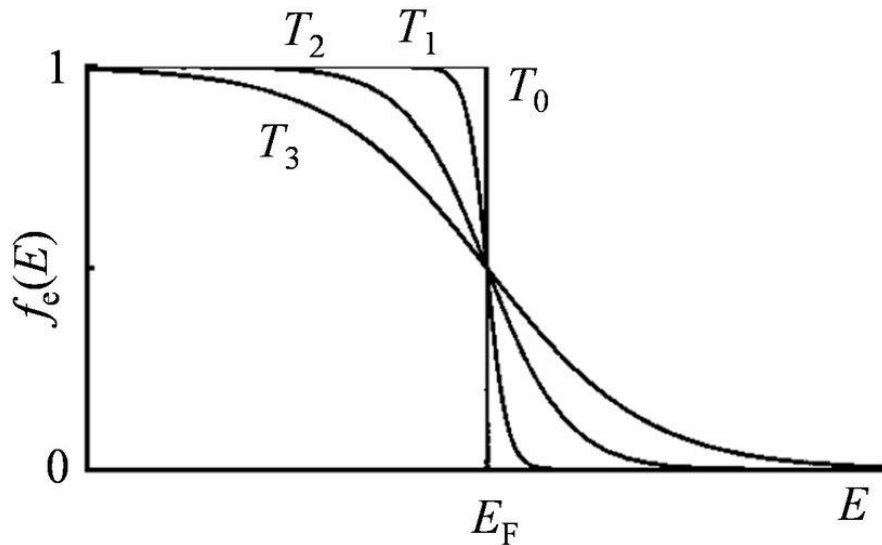


Fig : Fermi Dirac distribution

Chapter 2

Free Electron Theory Of Conduction In Metal

Crystalline Structure

A crystalline solid is a solid in which atoms bond with each other in a regular pattern to form a periodic collection or array of atoms. All crystals have periodicity, which leads to long-range order. The periodic array in which the atoms are located is called the crystal lattice.

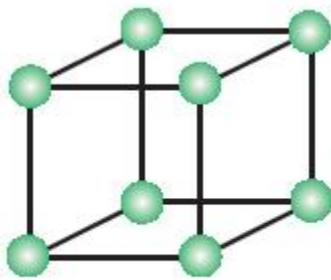
Before studying the crystalline structure we need to be clear about few terms.

- A. Unit Cell: A unit cell is the smallest representation of an entire crystal. All crystals lattices are built of repeating unit cells.
- B. Co-ordination Number: In a unit cell, an atom's co-ordination number is the number of atoms, it is touching.
- C. Atomic radius: It is half of the distance between the neighboring atoms in a crystal.

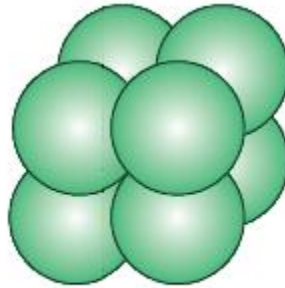
Following are three crystalline structure.

i. **Simple Cubic Structure**

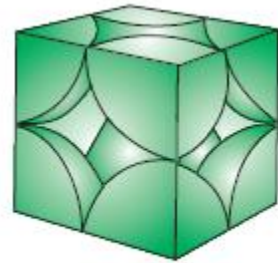
In a simple cubic structure is as follows.



(a)



(b)



(c)

Consider a cube of side 'a' and in SCC structure, an atom is present at each corners.

$$a = 2R$$

$$\text{No. of atoms per unit cell} = \frac{1}{8} \times 8 = 1$$

$$\text{Volume of unit cell} = a^3 = 8R^3$$

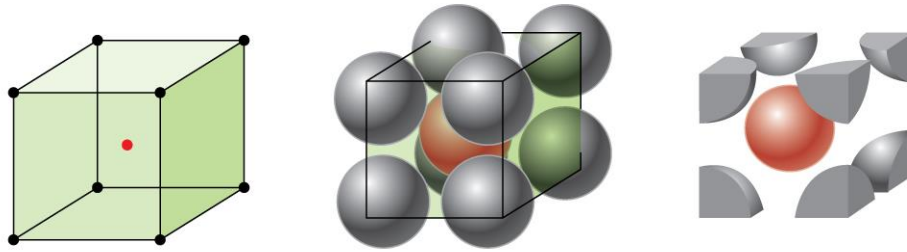
$$\text{Volume of an atom} = \frac{4}{3}\pi R^3$$

$$\text{Packing density} = \frac{\text{no of atom per unit cell} \times \text{Vol. of an atom}}{\text{Vol. of a unit cell}}$$

$$\text{Packing density} = \frac{1 \times \frac{4}{3}\pi R^3}{8R^3}$$

$$PD = 0.52$$

ii. **Body Centered Cubic Structure**



Body-centered cubic structure

Consider a cube of side 'a'. In BCC structure, an atom is present at the center of the body and an atom is present at each corner of the cube.

$$\text{Atomic radius} = R$$

$$\text{Body diagonal}(BD) = 4R$$

$$\text{Face diagonal}(FD) = \sqrt{2}a$$

$$(BD)^2 = (FD)^2 + a^2$$

$$(4R)^2 = 2a^2 + a^2$$

$$a^2 = \frac{16R^2}{3}$$

$$a = \frac{4R}{\sqrt{3}}$$

Now,

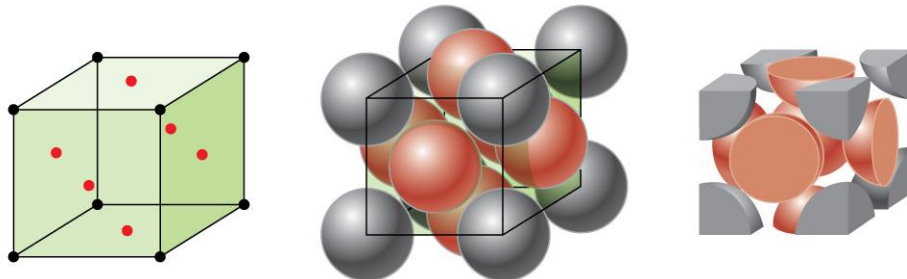
$$\text{No. of atoms per unit cell} = 1 + \frac{1}{8} \times 8 = 2$$

$$\text{Packing density} = \frac{\text{no of atom per unit cell} \times \text{Vol. of an atom}}{\text{Vol. of a unit cell}}$$

$$\text{Packing density} = \frac{2 \times \frac{4}{3}\pi R^3}{\left(\frac{4R}{\sqrt{3}}\right)^3}$$

$$PD = 0.6801$$

iii. **Face Centered Cubic Structure**



Face-centered cubic structure

Consider a cube of side 'a'. In FCC structure, an atom is present at each faces of the cube and an atom is present at each corner of the cube.

$$\text{Face diagonal}(FD) = 4R$$

$$\text{Side}(a) = \frac{4R}{\sqrt{2}}$$

$$\text{Body diagonal (BD)} = \left[(4R)^2 + \left(\frac{4R}{\sqrt{2}} \right)^2 \right]^{\frac{1}{2}} = 2\sqrt{6}R$$

$$\text{No. of atoms per unit cell} = \frac{1}{2} \times 6 + \frac{1}{8} \times 8 = 4$$

$$\text{Packing density} = \frac{\text{no of atom per unit cell} \times \text{Vol. of an atom}}{\text{Vol. of a unit cell}}$$

$$\text{Packing density} = \frac{\frac{4 \times \frac{4}{3} \pi R^3}{\left(\frac{4R}{\sqrt{2}} \right)^3}}$$

$$PD = 0.74$$

Lattice Constant

Lattice constant or lattice parameter refers to the physical dimension of unit cell in a crystal lattice. In case of cubic crystal structure, lattice constants are equal and we refer to 'a'.

Effective Mass Of An Electron

A particle's effective mass is the mass that it seems to have when responding to forces, or the mass that it seems to have with other particle in thermal distribution.

Kinetic energy of free particle is given as,

$$E = \frac{mv^2}{2}$$

$$E = \frac{m^2 v^2}{2m}$$

$$E = \frac{p^2}{2m}$$

$$E = \frac{\hbar^2 k^2}{2m} \quad [p = \hbar k \text{ where, } \hbar = \frac{h}{2\pi} \text{ and } k = \frac{2\pi}{\lambda}]$$

Differentiating above equation w.r.t k then,

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m}$$

$$\frac{1}{\hbar} \frac{dE}{dk} = \frac{\hbar}{m} k \text{ ----- (i)}$$

$$\text{Also, } \lambda = \frac{h}{p} = \frac{h}{mv}$$

$$v = \frac{h}{m\lambda}$$

$$v = \frac{\hbar k}{m} \text{ ----- (ii)}$$

From (i) and (ii) we get,

$$\frac{1}{\hbar} \frac{dE}{dk} = v \text{ ----- (iii)}$$

Also, acceleration of particle

$$a = \frac{dv}{dt}$$

$$a = \frac{1}{\hbar} \frac{d}{dt} \frac{dE}{dk} \quad [\text{using relation (iii)}]$$

$$\frac{dv}{dt} = \frac{1}{\hbar} \frac{d}{dt} \frac{dE}{dk}$$

$$\frac{d}{dt} \left(\frac{\hbar k}{m} \right) = \frac{1}{\hbar} \left\{ \frac{d}{dk} \frac{dk}{dt} \frac{dE}{dk} \right\}$$

$$\frac{\hbar}{m} \frac{dk}{dt} = \frac{1}{\hbar} \left\{ \frac{dk}{dt} \frac{d^2 E}{dk^2} \right\}$$

$$\frac{\hbar^2}{m} = \frac{d^2 E}{dk^2}$$

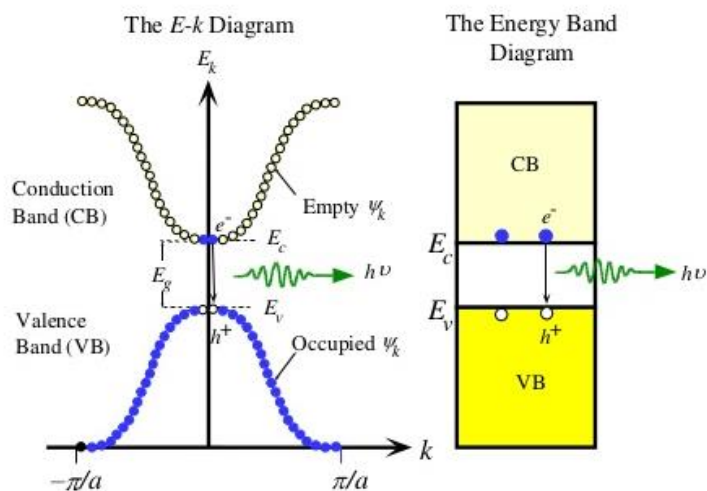
$$m = \frac{\hbar^2}{\frac{d^2 E}{dk^2}}$$

So, mass depends upon curvature of energy in k-space (E-k diagram).

For solid, the above expression of mass is known as effective mass and denoted by m^* .

$$\text{Therefore, } m^* = \frac{\hbar^2}{\frac{d^2 E}{dk^2}}$$

- Effective mass of electron is positive for $\frac{d^2 E}{dk^2} > 0$ i.e. Curvature is concave upward.
- Effective mass of electron is negative for $\frac{d^2 E}{dk^2} < 0$ i.e. Curvature is concave downward.
- For certain values of k the curvature is zero pointing towards the infinite mass.
- Mass is positive and electron is accelerated in the direction of applied external electric field.
- Mass is negative and electron is decelerated due to the external applied field.



Thermal Velocity Of Electron At Equilibrium

For N molecules of gas in a container, each molecule has its random velocity, change in momentum following collision with the wall of container. Change in momentum of a particle along positive x-direction after suffering collision is,

$$\Delta p_x = 2mv_x \text{ ----- (1)}$$

[v_x is the velocity of gas molecules in the positive x-direction and m is mass of gas molecule]

$$\text{Time interval } \Delta t = \frac{2a}{v_x}$$

Where, 'a' is the side of container.

$$\text{force on wall} = \frac{\Delta p_x}{\Delta t}$$

$$\text{force on wall} = \frac{mv_x^2}{a}$$

$$\text{Pressure on wall (P)} = \frac{\text{total force due to N-particle}}{\text{Area of face}}$$

$$\text{Pressure on wall (P)} = \frac{mv_{x1}^2 + mv_{x2}^2 + mv_{x3}^2 + \dots + mv_{xn}^2}{a \cdot a^2}$$

$$\text{Pressure on wall (P)} = \frac{mNv_x^2}{a^3} \left[v_x^2 = \frac{v_{x1}^2 + v_{x2}^2 + v_{x3}^2 + \dots + v_{xn}^2}{N} \right]$$

Since, motion is in 3-D

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

Assuming $v_x = v_y = v_z$

$$v^2 = 3v_x^2$$

$$v_x^2 = \frac{v^2}{3}$$

$$\text{Pressure on wall (P)} = \frac{Nm v^2}{3a^3} = \frac{Nm v^2}{3V} \text{ [Where, V= volume of container which is a cube]}$$

$$PV = \frac{Nm v^2}{3}$$

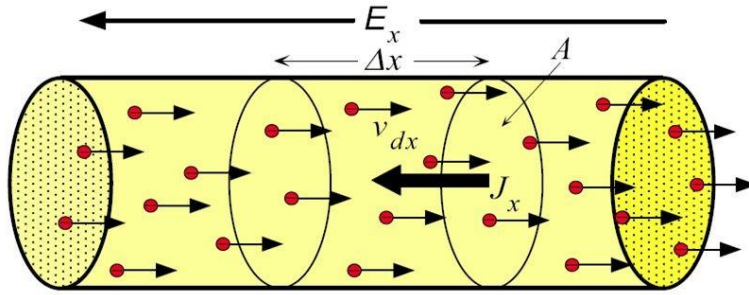
Average thermal energy per degree of freedom is $\frac{kT}{2}$. For 3-degree of freedom, it is $\frac{3kT}{2}$. Kinetic energy

(KE) per molecule is $\frac{mv^2}{2}$.

$$\text{So, } KE = \frac{1}{2}mv^2 = \frac{3kT}{2}$$

$$v_{th} = \left(\frac{3kT}{m} \right)^{\frac{1}{2}} \text{ [Where, } v_{th} \text{ is mean thermal velocity.]}$$

Electron Mobility, Conductivity And Resistivity



Due to random thermal motion no net current is detected. So to maintain the flow of electron electric field is applied. If V is the applied field across length then electric field is given by,

$$E = \frac{V}{L}$$

In this field, force experienced by an electron is given by,

$$F = -eE$$

Also, $F = m^* \times a$

$$a = \frac{-eE}{m^*}$$

Due to electric field electron gets velocity in opposite direction to applied field called drift velocity. The average time between two collision is termed as mean free time or relaxation time denoted by τ . If v_{av} is average velocity of electron then,

$$v_{av} = a\tau$$

Also, the average velocity v_{av} is called drift velocity. So, neglecting negative sign we get. (since electron drift in opposite direction of applied field)

$$v_D = a\tau$$

$$v_D = \frac{eE}{m^*} \tau$$

$$v_D = \mu_D E$$

Thus, $\mu_D = \frac{e\tau}{m}$ is called as drift mobility.

If all the electrons drift with velocity v_D , then number of electrons crossing plane of unit area per second is,

$$\text{Electron flow} = \frac{\text{Number of electrons}}{\text{Area} \times \text{time}}$$

$$\text{Electron flow} = \frac{\text{Number of electrons} \times \text{length}}{\text{Area} \times \text{length} \times \text{time}}$$

$$\text{Electron flow} = \frac{\text{Number of electrons}}{\text{Area} \times \text{length}} \times \frac{\text{length}}{\text{time}}$$

$$\text{Electron flow} = N_e \times \text{velocity}$$

$$\text{Electron flow} = N_e \times v_D$$

Where, N_e is electron density.

$$\text{Current density}(J) = \text{charge} \times \text{electron flow} = eN_e v_D$$

Also, we know $v_D = \mu_D E$

Therefore, $J = eN_e \mu_D E$

Again we have, $J = \sigma E$

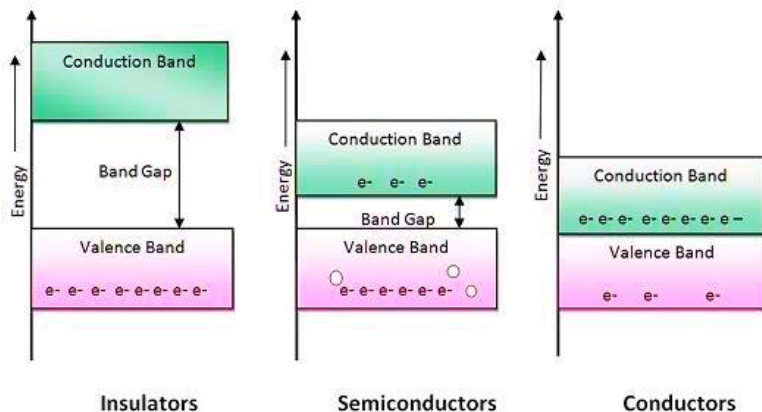
Where, σ is conductivity and E is applied electric field.

Also, $\text{Conductivity}(\sigma) = eN_e \mu_D$ and $\text{Resistivity}(\rho) = \frac{1}{\sigma} = \frac{1}{eN_e \mu_D}$

$$\text{Resistance}(R) = \frac{\rho l}{A}$$

Band Theory Of Solids

A useful way to visualize the difference between conductors, insulators and semiconductors is to plot the energies for electrons in the materials. Instead of having discrete energies as in the case of free atoms, the available energy states form bands. Crucial to the conduction process is whether or not there are electrons in the valence band or separated by a large gap from the conduction band, in conductors like metals. The valence band overlaps the conduction band and in semiconductor there is a small enough gap between the valence and conduction band that thermal or other excitation can bridge the gap with such a small gap, the presence of small percentage of a doping material can increase conduction dramatically. Energy bands in solids (Conductor, Semiconductor and insulator)

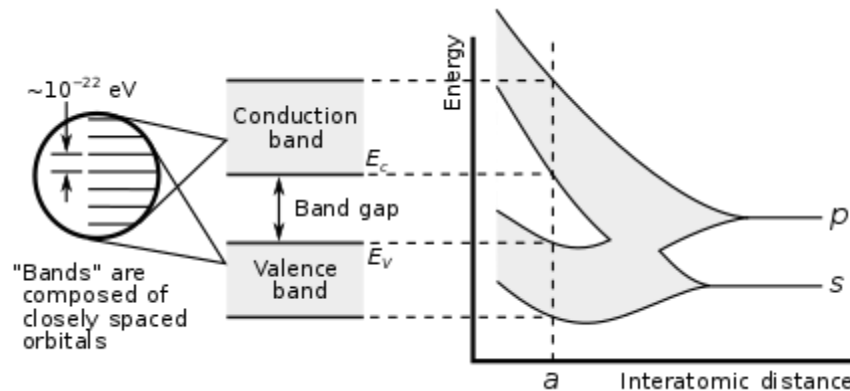


Energy Band Formation In Conductors

Let us take an example of Li atoms.

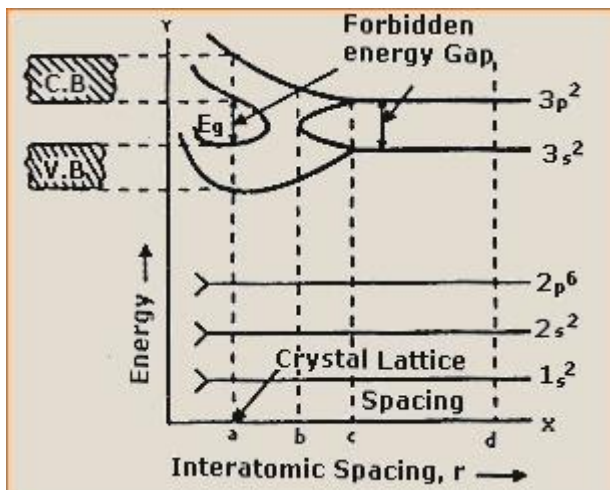
The spdf configuration of Li atoms are $1s^2 2s^1$. i.e. The 1s orbital is completely filled and 2s orbital is half filled. The valence electron has energy level of 2s. Consider 'N' Li atoms and because of close proximity of atoms there is interaction in the energy levels of electrons and thus the energy level start splitting forming bands.

We know 2s state can have 2 electrons and 2p state can have 6 electrons and thus they can have 2N and 6N states respectively. These energy levels split forming 2s band, 2p band and so on. The splitting of band gives rise to the valence and conduction band. Those states which are occupied by electrons at normal state from valence band and the excited states from conduction band.



Energy Band Formation In Semi-conductors

Let us take an example of Si atoms. The spdf configuration of Si atoms are $1s^2 2s^2 2p^6 3s^2 3p^2$. Here, 3s state have 2 electrons and thus it will have 2N states (considering N number of Si atoms in the system) and 3p state have 2 electrons and thus will have 2N states which are applied and 4N states will be empty. Since, 2p state can accommodate 6 electrons. Because of the close proximity of Si atoms there is interaction between the neighboring atoms and thus the energy level start splitting forming valence and conduction band.



Chapter 3

Conduction In Liquid And Gases

Ionic Conduction In Liquid

In liquid, free electron cannot move directly between anode and cathode. Here, only ionic movement is possible. For this to happen, there should be significant numbers of positive and negative ions formed by adding certain form of electrolyte into pure water. The ionic conductivity in liquid is,

$$\sigma_{ion} = eN_{ion}\mu_{ion}$$

Where, N_{ion} = Number of ions moving due to influence of external field

μ_{ion} = mobility of ions

During movements of ions, they collide and exchange energy from Einstein diffusion relation.

$$\mu_{ion} = \frac{eD}{kT}$$

Where, D= Diffusion Coefficient for ions.

k= Boltzmann's constant

Diffusion coefficient varies with temperature,

$$D = D_0 e^{-Q/kT}$$

Where, D_0 = the pre-exponential factor

Q= activation energy for diffusion

Now,

$$\sigma_{ion} = eN_{ion} \frac{eD}{kT}$$

$$\sigma_{ion} = N_{ion} \frac{e^2}{kT} D_0 e^{-Q/kT}$$

$$\sigma_{ion} = \sigma_0 e^{-Q/kT}$$

Where,

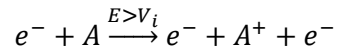
$$\mu_0 = \frac{e^2 N_{ion}}{kT} D_0$$

Taking ln on both side,

$$\ln \sigma_{ion} = \ln \sigma_0 - \frac{Q}{kT}$$

Electrical Conduction In Gases

Gas is perfect insulator during room temperature and normal state. When external field and other additional energies are applied across the gas, then free electrons may start to liberate with avalanche phenomenon as below.



Where,

E= energy gained by electron during collision

V_i =Ionization potential

A is atom, A^{+} is positive ion and e^{-} is electron.

A few electrons are produced by UV rays falling on the cathode, ionize neutral gas particle producing positive ions and additional electrons. These additional electrons make ionizing collision. The positive ion reach the cathode and give rise to secondary electron.

Let n_0 be the number of electrons emitted from cathode, α is ionization coefficient and n_x be the number of electrons at a distance x from the cathode.

At $x=0$, $n_x=n_0$

Similarly, $\frac{dn_x}{dx} = \alpha n_x$

$$n_x = n_0 e^{\alpha x}$$

Number of electron reaching the anode,

$$x = d$$

$$n_d = n_0 e^{\alpha d}$$

Current in the gap is equal to number of electrons travelling per second.

$$I = I_0 e^{\alpha d}$$

Where I_0 = initial current at cathode.

Arc Discharge And Electrical Breakdown In Gas

In normal gaseous condition, electron avalanche process occur. In this avalanche process, positive ion of the gas molecule may liberate free electron from cathode. When such liberated electron impinge on gaseous molecule, photo emission phenomenon also occur such photon energy may help to generate other free electrons. As liberation of electron from cathode terminal was primary emission of free electron, additional electron generated from photon is called secondary emission process.

$$\text{i.e. } n_0'' = n_0 + n_0'$$

n_0 = number of free electron available in gas

n'_0 = Number of electron generated from emission

n''_0 = total number of electrons leaving from cathode

Then, total number of electron reaching from cathode to anode is,

$$n = n''_0 e^{\alpha d}$$

Where,

n= total number of electron reaching from cathode to anode

d= distance between cathode and anode

α = ionization coefficient

Streaming

A single electron starting at the cathode due to ionization of gaseous molecules build up on avalanche phenomena. The electron in avalanche may move very fast compared with positive ion. Thus, electron with primary and secondary emission phenomena move with very high speed and reaches up to anode. A space charge again will be created between cathode and anode from where electrons can flow in significant amount. This situation is called streaming phenomena.

Arc Discharge

If the current in the gap is increased above I_A (critical current), the voltage across the gap suddenly reduce to few volt. The discharge becomes very luminous and noisy called arc discharge.

Chapter 4

Dielectric Material

We know, conductors are required for the conduction of current whereas dielectric are required to have insulation at various high voltage application and unwanted content. A dielectric material is an electrical insulator that can be polarized by an applied electric field.

When a dielectric is placed in an electric field, electric charges do not flow through the material as they do in a conductor but they slightly shift from their average equilibrium positions causing dielectric polarization. Because of dielectric polarization, positive charges are displaced toward the field and negative charge shift in opposite direction.

Parallel Plate Capacitor

$$C = \frac{\epsilon_0 A}{d}$$

Where,

ϵ_0 is absolute permittivity

A is plate area

d is distance of separation

If there is a material medium between the plates, then capacitance increases by a factor ϵ_r is called the dielectric constant of the medium or relative permittivity.

The increase in capacitance is due to the polarization of the medium in which positive and negative charges are displaced with respect to their equilibrium position. The opposite surfaces of the dielectric medium acquire opposite surface charge densities that are related to the amount of polarization in the material.

Electric dipole moment P, which is measure of electrostatic effects of a pair of opposite charges +Q and -Q separated by a finite distance 'a',

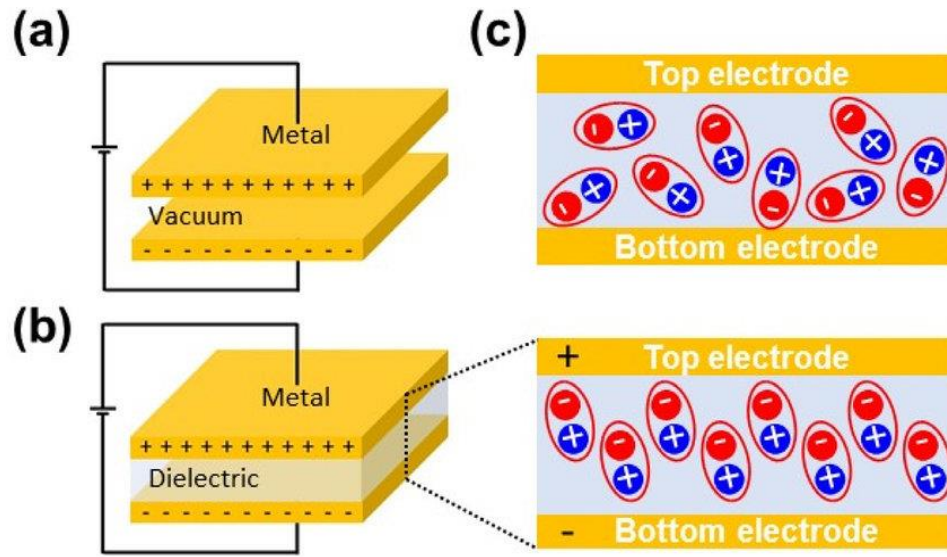
$$P = Qa$$

Electric susceptibility (χ_e) : Electric susceptibility of dielectric is the measure of how easily it polarizes in response to an electric field.

$$\chi_e = \frac{\text{Polarization (P)}}{\epsilon_0 \times \text{electric field (E)}}$$

$$\text{Relative permittivity}(\epsilon_r) = \chi + 1$$

Macroscopic Approach Of Permittivity



Case (a) : Parallel plate capacitor without dielectric material

Case (b) : Parallel plate capacitor with dielectric material

For case (a) :

$$\text{Capacitance } (C_0) = \frac{Q_0}{V}$$

After insertion of dielectric between the plates, then the charges on the surface of plate get increased due to appearance of the charge on the surface of the dielectric near to the capacitor plates.

For case (b):

$$\text{Capacitance } (C) = \frac{Q}{V}$$

From case (a)

$$C_0 = \frac{Q_0}{V}$$

$$C_0 V = Q_0$$

$$V \times \frac{\epsilon_0 A}{d} = Q_0$$

$$\frac{\epsilon_0 V}{d} = \frac{Q_0}{A} = \sigma_0$$

From case (b)

$$C = \frac{Q}{V}$$

$$CV = Q$$

$$V \times \epsilon \frac{A}{d} = Q$$

$$\frac{V\epsilon}{d} = \frac{Q}{A} = \sigma$$

Where, σ = surface charge density

Now, from above equations

$$\frac{\sigma}{\sigma_0} = \frac{Q}{Q_0} = \frac{C}{C_0} = \epsilon_r$$

Therefore, dielectric constant or relative permittivity is defined as the increase in charge stored on capacitor plates due to insertion of dielectric medium compared to the charge without dielectric medium.

Increase in surface charge density when dielectric medium is inserted compared with that without dielectric is called polarization.

$$Polarization(P) = \sigma - \sigma_0 = \frac{Q}{A} - \frac{Q_0}{A}$$

$$P = \frac{\epsilon_0 \epsilon_r V}{d} - \frac{\epsilon_0 V}{d}$$

$$P = \frac{\epsilon_0 V}{d} (\epsilon_r - 1)$$

$$P = \epsilon_0 \chi_e E$$

Polarizability is the relative tendency of a charge distribution like the electron cloud of an atom or molecule to be distorted from its normal shape by an external field.

Polarizability α_e is the ratio of induced dipole moment (P) of an atom to electric field (E).

$$\alpha_e = \frac{Z_e^2}{m_e^* \omega_0^2}$$

Where,

e is electronic charge

ω_0 is oscillation frequency of center of electron cloud around nucleus

Z is an atomic number and m_e^* is effective mass of electron.

Clausius Massotti Equation

The actual field experienced by a molecule in dielectric is defined as local field (E_{LOC}) which depends not only on the free charges on the plates but also on the arrangement of all the polarized molecules around this point. The greater the polarization the greater the local field is given by,

$$E_{LOC} = E + \frac{P}{3\epsilon_0}$$

Induced polarization of molecule/atom is given by,

$$P_{ind} = \alpha_e E_{LOC}$$

Now, polarization of whole material is sum of induced polarization of each atom.

$$Polarization(P) = NP_{ind}$$

$$P = N\alpha_e E_{LOC}$$

$$P = N\alpha_e \left(E + \frac{P}{3\epsilon_0} \right)$$

$$P = N\alpha_e E + \frac{N\alpha_e P}{3\epsilon_0}$$

$$P - \frac{N\alpha_e P}{3\epsilon_0} = N\alpha_e E$$

$$P = \frac{N\alpha_e E}{\left(1 - \frac{N\alpha_e}{3\epsilon_0}\right)} \text{ --- (i)}$$

Also, we have

$$P = \epsilon_0 \chi_e E$$

$$\frac{N\alpha_e E}{\left(1 - \frac{N\alpha_e}{3\epsilon_0}\right)} = \epsilon_0 \chi_e E \text{ (from i)}$$

$$\epsilon_0 \chi_e = \frac{N\alpha_e}{\left(1 - \frac{N\alpha_e}{3\epsilon_0}\right)}$$

$$\epsilon_0 \chi_e \left(1 - \frac{N\alpha_e}{3\epsilon_0}\right) = N\alpha_e$$

$$\epsilon_0 \chi_e - \chi_e \frac{N\alpha_e}{3} = N\alpha_e$$

$$\epsilon_0 \chi_e = N\alpha_e + \chi_e \frac{N\alpha_e}{3}$$

$$\epsilon_0 \chi_e = \frac{N\alpha_e}{3} (\chi_e + 3)$$

$$\frac{\chi_e}{(\chi_e + 3)} = \frac{N\alpha_e}{3\epsilon_0}$$

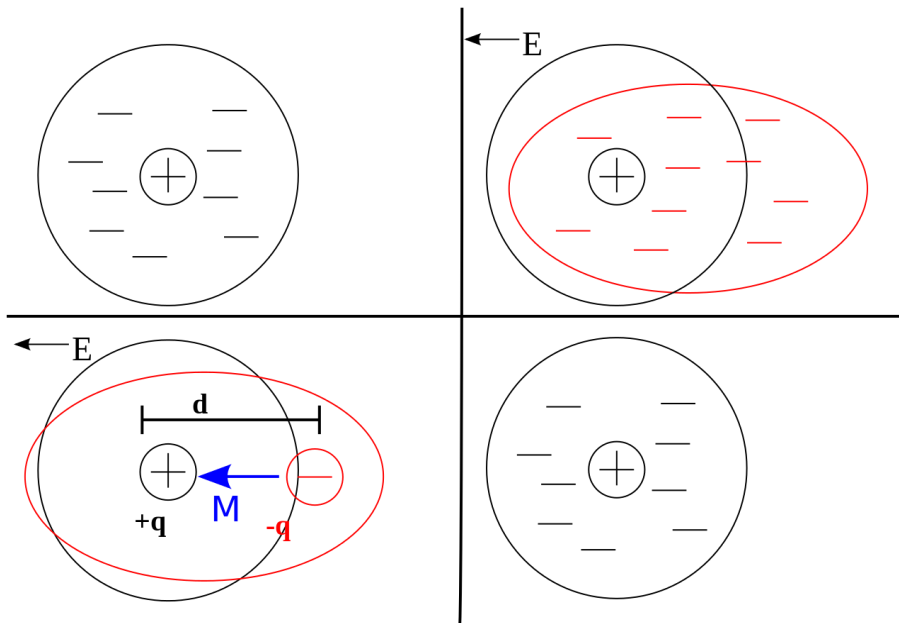
$$\frac{(\epsilon_r - 1)}{(\epsilon_r + 2)} = \frac{N\alpha_e}{3\epsilon_0}$$

The above equation is known as Clausius-Massotti equation. It relates relative permittivity with electronic polarizability.

Types of Polarization

- i. Electronic Polarization or Dielectric Polarization
- ii. Ionic Polarization
- iii. Oriental/Dipolar Polarization
- iv. Interfacial Polarization

I. Electronic Polarization



A material is made up of atoms. Each atom consists of a cloud of negative charge (electrons) bound to and surrounding a positive point charge at its center. In the presence of an electric field the charge cloud is distorted as shown in the top right figure.

This can be reduced to simple dipole using superposition theorem. A dipole is characterized by its dipole moment. A dipole moment is a vector quantity pointing from negative to positive charge.

$$M = qd$$

II. Ionic Polarization

In this case, a solid material must have some ionic character. It then automatically has internal dipoles but these built in dipoles cancel each other.

Let us consider a simple ionic crystal eg. NaCl. The lattice can be considered to consist of Na^+ and Cl^- dipoles as shown in the diagram.

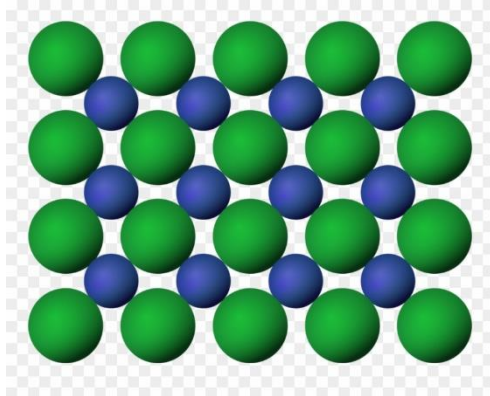


Fig: Greens are Cl^- and blues are Na^+

Each $\text{Na}^+ - \text{Cl}^-$ pair is a natural dipole, no matter how you pair up two atoms. Polarization of a given volume, however, is exactly zero because for every dipole moment, there is a neighboring one with exactly the same magnitude but opposite sign.

Note: The dipoles cannot rotate, their direction is fixed.

In presence of electric field, ions feel forces in opposite directions. The lattice distorts a little bit. Na^+ ions move a little bit towards the direction of applied field and Cl^- ions to opposite direction. The dipole moments between adjacent NaCl pairs in field direction are now different and there is net dipole moment in a finite volume.

$$P_{av} = \alpha_i E_{LOC}$$

α_i is ionic polarizability which is greater than α_e generally, $\alpha_i > 10\alpha_e$.

$$P = N_i \alpha_i E_{LOC}$$

N_i is number of ions.

$$E_{LOC} = E + \frac{P}{3\epsilon_0}$$

$$P = \epsilon_0 \chi_e E$$

We obtain,

$$\frac{(\epsilon_r - 1)}{(\epsilon_r + 2)} = \frac{N\alpha_e}{3\epsilon_0}$$

Electronic polarization is also present in ionic solids but their contribution is negligible compared to ionic polarization.

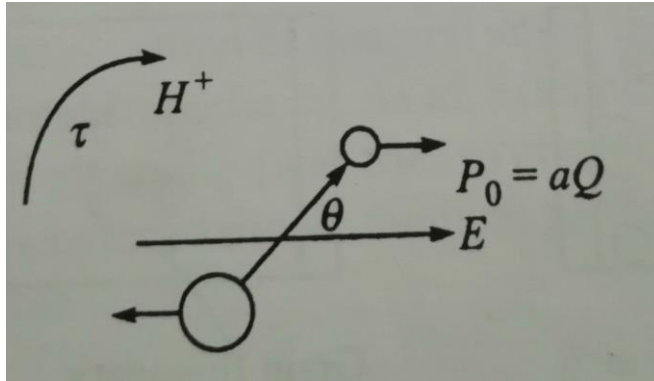
III. Orientational Polarization / Dipolar Polarization

This type of polarization is observed in liquid and gas. In these material natural dipoles exist which are free to rotate. In thermal equilibrium, the dipoles will be randomly oriented and thus carry no net polarization. The external field aligns these dipoles to some extent and thus induces a polarization of the material.

The prime example is liquid HCl, where every HCl molecule is little dipole that can have any orientation with respect to other molecules. Moreover, the orientation changes all the time because the molecules move. Therefore, in absence of external field net dipole moment is zero.

If we introduce electric field, dipoles would have tendency to turn into field because that would lower their energy. In reality, orientation of dipole and the field direction would be counter balanced by random collision with other dipoles and this process is energized by the thermal energy 'kT'.

So, the orientation of dipole is little bit shifted so that an average orientation in the field direction results.



Torque experienced by a dipole $(\tau) = F \sin \theta \times a$

$$\tau = EQa \sin \theta$$

$$\tau = EP_0 \sin \theta$$

Where, *dipole moment* $(P_0) = Qa$

If *potential energy* $(PE) = 0$ at $\theta = 0^\circ$ and $PE = PE_{max}$ at $\theta = 180^\circ$ then,

$$PE_{max} = \int_0^{180} E \sin \theta P_0 d\theta$$

$$PE_{max} = EP_0 \int_0^{180} \sin \theta d\theta$$

$$PE_{max} = EP_0 [-\cos \theta]_0^{180}$$

$$PE_{max} = EP_0 [-\cos 180 + \cos 0]$$

$$PE_{max} = 2EP_0$$

$$\text{Average dipole potential energy} = \frac{PE_{max}}{2}$$

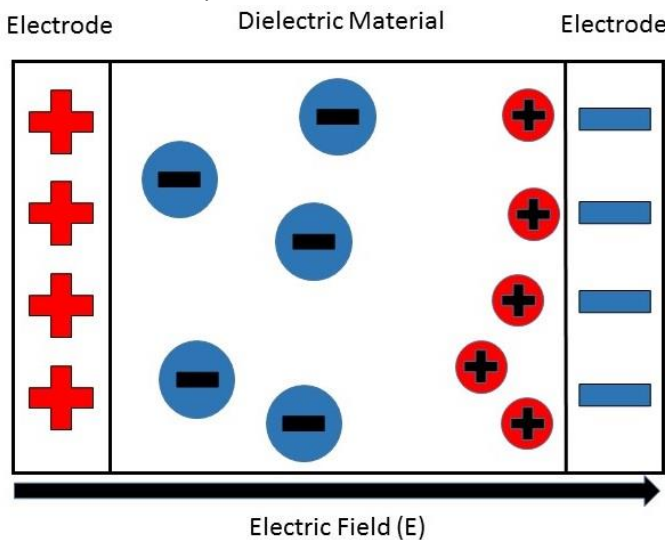
$$\text{Average dipole energy} \propto P_0 \times \frac{\text{Average dipole potential energy}}{\text{Average thermal energy}}$$

$$\text{Average dipole energy} \propto P_0 \times \frac{EP_0}{\frac{5kT}{2}}$$

Using Boltzmann statistics, average dipole moment per molecule = $\frac{P_0^2 E}{3kT} = \alpha_d E$
 Where, α_d = dipolar or orientational polarizability.

IV. Interfacial Polarization

Interfacial or space charge polarization occurs when there is an accumulation of charge at an interface between two materials or between two regions within a material because of an external field. This can occur when there is a compound dielectric, or when there are two electrodes connected to a dielectric material. This type of electric polarization is different from orientational and ionic polarization because instead of affecting bound positive and negative charges i.e. ionic and covalent bonded structures, interfacial polarization also affects free charges as well. As a result interfacial polarization is usually observed in amorphous or polycrystalline solids. Figure below shows an example of how free charges can accumulate in a field, causing interfacial polarization. The electric field will cause a charge imbalance because of the dielectric material's insulating properties. However, the mobile charges in the dielectric will migrate over maintain charge neutrality. This then causes interfacial polarization.



Since interfacial polarization occurs at the interface, so it cannot be added linearly with other type of polarization. So, under the influence of electronic, ionic and orientational polarization, induced dipole moment per molecule/atom

$$= \alpha_e E_{LOC} + \alpha_i E_{LOC} + \alpha_d E_{LOC}$$

Dielectric breakdown in solids

Dielectric breakdown is the long reduction in the resistance of an electrical insulator in the resistance of an electrical insulator when the voltage applied across it exceeds the breakdown voltage. This results in the electrical insulator becoming electrically conductive.

Maximum electric field that can be applied to the insulating medium without causing dielectric breakdown is called dielectric strength.

Dielectric strength depends on impurities on material, sample geometry, temperature, ambient conditions, duration and frequency of applied field, nature of electrodes etc.

Mechanism of Dielectric Breakdown in Solid

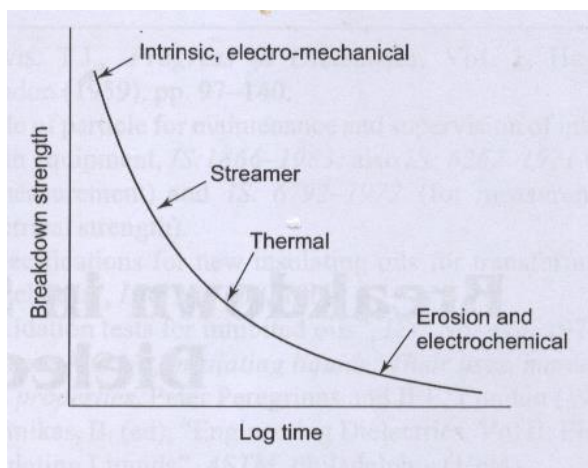


Fig. Variation of Breakdown Strength

I. Intrinsic Breakdown

When large voltage is applied for short duration, intrinsic breakdown takes place.

Intrinsic breakdown depends upon the presence of free electrons which are capable of migration through the lattice of dielectric.

Usually small number of conduction electrons are present in solid dielectric, these electrons get multiplied with the application of high field.

When an electric field is applied electrons gain energy from electric field and cross the band gap to reach conduction band.

This process is repeated, more and more electrons are available in conduction band eventually leading to breakdown.

II. Electromechanical Breakdown

When solid dielectrics are subjected to high electric fields, failure occurs due to electrostatic compressive force strength if it exceeds mechanical withstand of the material.

Compressive force $\propto V^2$

When the material is subjected to high stress theory of elasticity does not hold good and plastic deformation has to be considered.

Mechanical compression may also cause thermal breakdown, shearing of internal cracks, crack propagation and eventually dielectric failure.

III. Thermal Breakdown

When an electric field is applied to a dielectric conduction current flows through the material. (However it may be small)

(Joule heating σE^2 or dielectric loss $V^2 \omega \tan \delta$)

The current heats up the specimen and temperature rises.

Heat generated is transformed to the surrounding medium by conduction through solid dielectric and by radiation from its outer surfaces.

When heat generated exceeds heat lost, the electrical conductivity of dielectric increases due to increase in temperature.

Many local hot spots may form permanent conduction channel leading to dielectric breakdown.

IV. Breakdown Due to Internal Storage

Solid insulating materials contain void or cavities within the medium or at the boundaries between the dielectric and the electrodes.

These voids have lower dielectric strength as compared to the material.

Hence even under the normal working voltage the field in the voids may exceed their breakdown value and breakdown may occur.

Breakdown channel spreads through the insulation in an irregular tree like fashion leading to the formation of conducting channel.

Ferroelectricity

- Certain crystal like Barium Titanate (BaTiO_3) which have permanent polarization even in the absence of an applied field called as ferroelectric crystals and the phenomenon ferroelectricity.
- The temperature above which the ferroelectricity is lost is called Curie temperature (T_c). The Curie temperature for BaTiO_3 is 130°C .
- Below Curie temperature the center of mass of positive and negative charges do not coincide and thus net polarization exist. But above Curie temperature the center of mass of positive and negative charges coincide and thus net polarization is zero.
- Below T_c whole specimen gets spontaneously polarized and above T_c , disorderedness of dipole increases which is called as pyroelectric state.
- Ferroelectric crystals have large dielectric constants.
- Above T_c , the dielectric constant usually follows Curie-Weiss behavior which is given by,

$$\epsilon_r = \frac{C}{T - T_0}$$

C = Curie constant

T₀ = Curie-Weiss temperature

- The variation of polarization with electric field is not linear for such crystals but forms a closed loop called ferromagnetic loop which is similar to hysteresis loop.
- The ferroelectric crystals may be classified in two main groups
 - I. Order-disorder group
 - II. Displacive group
- In order-disorder group, the ferroelectric transition is associated with individual ordering of ions. Eg. KH₂PO₄, RbH₂PO₄ etc.
- The displacive group of ferroelectric materials is one in which the ferroelectric transition is associated with displacement of sub-lattice of ions of one type relative to sub-lattice of another type.

Piezo-electricity

- Certain crystals become polarized in response to an applied mechanical stress. It is called direct piezo-electric effect.
- Materials exhibiting direct piezoelectric effect also exhibit the reverse piezoelectric effect. (i.e. The internal generation of mechanical strain resulting from applied electric field)
- The two effects are commonly known as piezo-electricity.
- Only certain crystals which have no center of symmetry exhibit piezo-electricity. In absence of external force center of mass of positive and negative charges coincide. But when external force is applied center of mass of positive and negative charges get shifted giving rise to net polarization.
- The polarization direction depends on the direction of applied force too.
- Similarly, an electric field applied to piezoelectric material results in the generation of mechanical strain and if the direction of applied field is reversed then the strain developed is also shifted in another direction.

Let T_j be applied mechanical stress in j direction. P_i is the induced polarization along i direction then,

$$P_i = d_{ij}T_j$$

d_{ij} is piezoelectric coefficient

Similarly, for reverse piezoelectric effect,

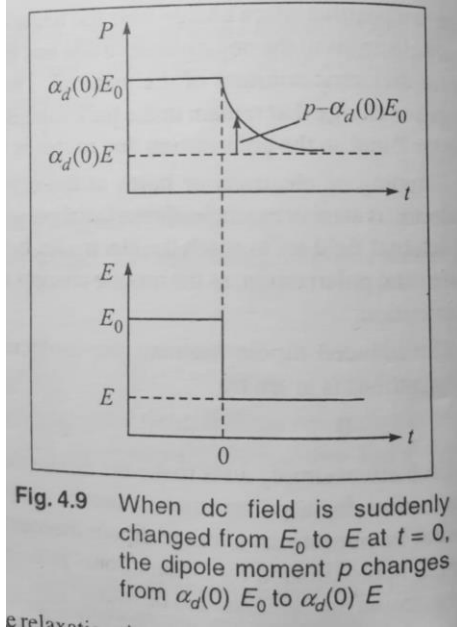
$$S_j = d_{ij}E_i$$

Where, S_j is induced strain along j direction.

E_i is applied electric field along i direction.

Dielectric losses, Frequency and Temperature

When a dielectric material is placed in an ac electric field, the field tries to polarize the material in different direction in each half cycle. But the thermal motion and intermolecular interactions try to randomize the orientation. So, the dipoles cannot respond instantaneously to the changes in field. At low frequency dipoles can respond to the change in field, so dipolar polarizability is maximum in the region. At high frequency, its value decreases.



Assume, τ is the average relaxation time between molecular collisions. If p is the induced dipole moment then $p - \alpha_d(0)E_0$ is the excess moment which must disappear as t tends to infinity. τ is the average time to eliminate the excess dipole moment. Rate of change of dipole moment is.

$$\frac{dp}{dt} = -\frac{p - \alpha_d(0)E}{\tau}$$

In case of ac, $E = E_0 \sin \omega t = E_0 e^{j\omega t}$

Then,

$$\begin{aligned} \frac{dp}{dt} &= \frac{\alpha_d(0)E}{\tau} - \frac{p}{\tau} \\ \frac{dp}{dt} &= \frac{\alpha_d(0)E_0 e^{j\omega t}}{\tau} - \frac{p}{\tau} \\ P &= \alpha_d E_0 e^{j\omega t} \end{aligned}$$

Where,

$$\alpha_d(\omega) = \frac{\alpha_d(0)}{1 + j\omega\tau}$$

Where, $\alpha_d(\omega)$ represents dipolar polarizability under ac conditions.

Thus, we observe that polarizability is the function of frequency. The ac dipolar polarizability of any liquid is complex, so we can obtain complex dielectric constant.

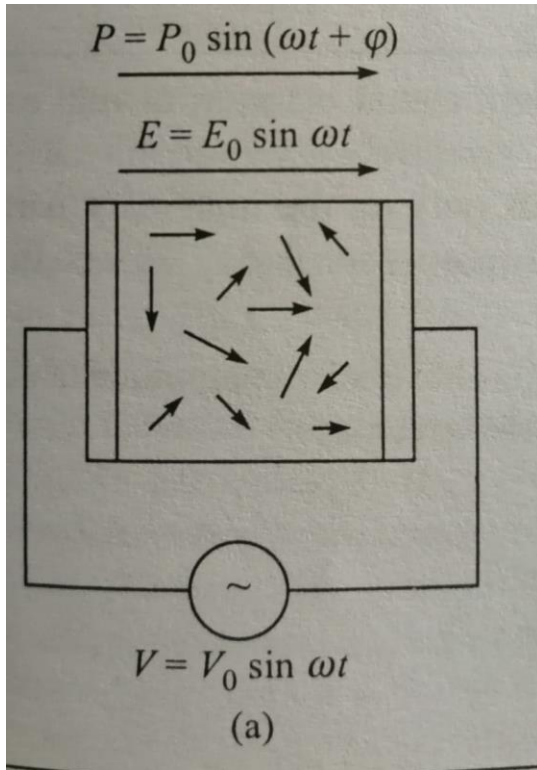
$$\epsilon_r = \epsilon'_r - j\epsilon''_r$$

Where,

ϵ'_r = real part of dielectric constant

ϵ''_r = imaginary part of dielectric constant

ϵ'_r and ϵ''_r are frequency dependent.



The capacitance of a capacitor is given by,

$$C = \frac{\epsilon_0 \epsilon_r(\omega) A}{d}$$

$$C = \frac{\epsilon_0 [\epsilon'_r(\omega) - j\epsilon''_r(\omega)] A}{d}$$

The capacitive susceptance $Y = j\omega C$

$$Y = \frac{j\omega\epsilon_0 [\epsilon'_r(\omega) - j\epsilon''_r(\omega)] A}{d}$$

$$Y = \frac{j\omega\epsilon_0 \epsilon'_r(\omega) A}{d} + \frac{\omega\epsilon_0 \epsilon''_r(\omega) A}{d}$$

$$Y = j\omega C + G_p$$

Where,

$$C = \frac{\epsilon_0 \epsilon_r'(\omega) A}{d}$$

$$G_p = \frac{\omega \epsilon_0 \epsilon_r''(\omega) A}{d}$$

C depends on real part of dielectric constant G_p depends on imaginary part of dielectric constant. $G_p = 1/R_p$ is responsible for power loss in dielectric medium.

G_p is called conductance of the capacitor. The capacitor can be represented by an ideal capacitor with capacitance C and resistor with resistance $R = 1/G_p$ in parallel to represent the power loss in dielectric medium.

Total input power to dielectric (P_{in}) = VI

$$P_{in} = V \cdot VI$$

$$P_{in} = V^2(j\omega C + G_p)$$

$$P_{in} = V^2 j\omega C + G_p V^2$$

$$\text{Real power dissipation} = G_p V^2 = P_{Loss}$$

Loss tangent or loss factor is given by,

$$\tan \delta = \frac{\epsilon_r''}{\epsilon_r'}$$

$$\text{Power loss per unit volume, } P_{vol} = \frac{P_{Loss}}{V}$$

$$P_{vol} = \frac{G_p V^2}{Ad} = V^2 \frac{\omega \epsilon_0 \epsilon_r'' A}{d} \frac{1}{Ad} = \frac{V^2}{d^2} \omega \epsilon_0 \epsilon_r'' = E^2 \omega \epsilon_0 \epsilon_r' \tan \delta$$

The expression relates the dependence of dielectric loss on frequency.

$$\text{Also, } \alpha_d = \frac{1}{T}$$

So, at higher temperature polarization of material is randomized by increased thermal vibration. So, dielectric constant also decreases at higher temperature (since α_d is linearly related with ϵ_r). Therefore, dielectric strength of material is lower at higher temperature.

Chapter 5

Magnetic Materials

Introduction

Materials in which a state of magnetization can be induced are called magnetic materials. When magnetized, such materials create a magnetic field in the surrounding space. All the molecules of a material contain electrons orbiting around the nucleus.

These orbits are therefore equivalent to circulating currents and so develop an m.m.f. Depending on, whether any non-neutralized orbit exists or not, specimens can be said to be magnetized or unmagnetized.

The readiness of a material to accept magnetism is expressed by its permeability.

Permeability

Magnetic permeability (μ) is the ability of a magnetic material to support magnetic field development. In other words, magnetic permeability is the constant in the proportionality between magnetic induction and magnetic field intensity. The greater the magnetic permeability of the material, the greater the conductivity for magnetic lines of force, and vice versa. The magnetic permeability of a material indicates the ease with which an external magnetic field can create a higher magnetic force of attraction in the material. The SI unit of magnetic permeability is Henry per meter.

$$\mu = \frac{B}{H}$$

Relative Permeability

Relative permeability of the medium is defined as fractional increase in the field with respect to the field in free space when a material medium is introduced.

$$\mu_r = \frac{B}{B_0} = \frac{B}{\mu_0 H}$$

$$\mu = \mu_0 \mu_r$$

Susceptibility

Magnetic susceptibility is a dimensionless proportionality constant that indicates the degree of magnetization of a material in response to an applied magnetic field.

$$\chi = \frac{M}{H}$$

Where M denotes magnetic dipole moment per unit volume and H denotes magnetic field strength then magnetic induction B is given by,

$$B = B_0 + \mu_0 M$$

$$B = \mu_0 (H + \chi_m H)$$

$$B = \mu_0 (1 + \chi_m) H$$

$$B = \mu_0 \mu_r H$$

Where, $\mu_r = \chi_m + 1$

Types of Magnetism

1. Ferromagnetism

- These materials gets easily/strongly magnetized after application of magnetic field.
- Ferromagnetic materials can possess large permanent magnetization even in the absence of an applied field.
- Magnetic susceptibility (χ_m) is positive and very large and depends on the applied field intensity.
- The relation between magnetization M and $\mu_0 H$ is highly non-linear.
- Ferromagnetism occurs below a critical temperature (T_c) called Curie temperature and above this temperature ferromagnetism is lost.
- It follows Curie-Weiss law,

$$\chi = \frac{C}{T - T_c}$$

C is constant for the material

T is absolute temperature

T_c is curie temperature

- Example : Iron, nickel cobalt

2. Ferrimagnetism

- It is mainly seen in ferrites (oxides of iron).
- It is similar to that of ferromagnetism but has two distinctions.
 - I. Curie temperature is low for ferrimagnetic materials.
 - II. Alignment of magnetic domains.

Ferromagnetic 	Below T_C , spins are aligned parallel in magnetic domains
Antiferromagnetic 	Below T_N , spins are aligned antiparallel in magnetic domains
Ferrimagnetic 	Below T_C , spins are aligned antiparallel but do not cancel
Paramagnetic 	Spins are randomly oriented (any of the others above T_C or T_N)

3. Paramagnetism

- Paramagnetic materials have small positive magnetic susceptibility. ($\times 10^{-2}$ range)
- In the absence of external magnetic field, atomic moments (magnetic moments) are randomly oriented due to random collision of molecules. So, the average dipole moment and the net magnetization are both zero.
- But when an external field is applied they get feebly magnetized and is equal to $\chi_m H$.
- Degree of magnetization increases with increasing field. (i.e increase in $\mu_0 H$) and vice-versa.
- It follows Curie law given by,

$$\chi_m = \frac{C}{T}$$
- Example : Pt, Al, Cr, Mn and dilute solutions of ferromagnetic materials.

4. Diamagnetism

- These materials have negative magnetic susceptibility. A perfect diamagnet has $\chi_m = -1$.
- When diamagnetic materials are placed in magnetic field, the magnetization vector (M) is in opposite direction to the applied field. This causes the magnetic field within the material to be less than the applied field.
- The negative susceptibility can be interpreted as the diamagnetic substance trying to expel the applied field from within the material.
- Covalent crystals and many ionic crystals are diamagnetic.
- Examples: Bi, Sb, Au, Cu, Hg etc.

5. Anti-ferromagnetism

- Have small positive susceptibility.
- In absence of external field there is no net magnetization of the material.
- They possess magnetic ordering in which the magnetic moment on alternating atoms in the opposite directions.
- Antiferromagnetism exists below a critical temperature called Neel temperature.
- Above Neel temperature, they are paramagnetic.
- Example: MnO , FeO , NiO , FeCl_3 etc.

Domain Structure

When cooled below a temperature called curie temperature, the magnetization of a ferromagnetic material spontaneously divides into many small regions called magnetic domain.

A magnetic domain is the region within a magnetic material in which the magnetization is in uniform direction. This means that individual magnetic moments of the atoms are aligned with one another and they point in the same direction. The magnetization within each points in a uniform direction but the magnetization of different domain may point in different directions. Domain structure is responsible for ferromagnetism.

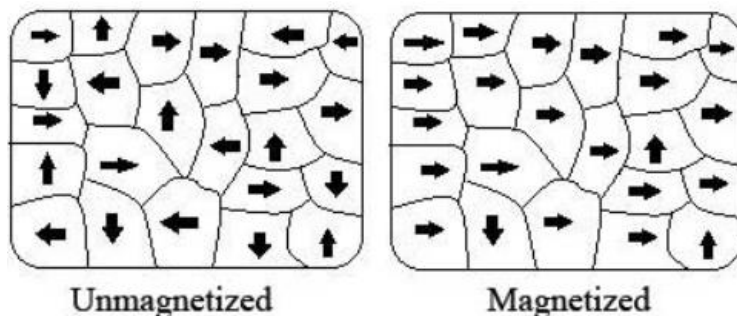


Fig: Domains in unmagnetized and magnetized iron

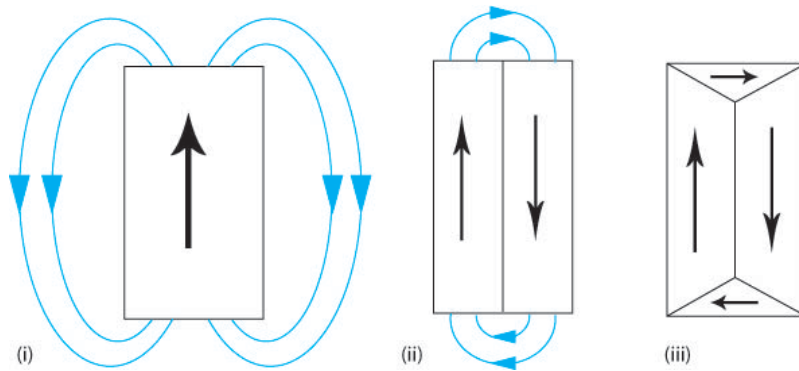
Why domains form?

The reason a piece of magnetic material such as iron spontaneously divides into separate domains rather than exist in a state with magnetization in the same direction throughout the material, is to minimize internal energy.

A large region of ferromagnetic materials with a constant magnetization throughout crystal will create a large magnetic field extending into the space outside itself. This requires a lot of energy stored in the field.

To reduce magnetostatic energy, sample can split into two domains, with the magnetization in opposite directions in each domain. To reduce the field energy further, each of these domains can split. Also, it

results in smaller parallel domains with magnetization in altering directions, with smaller amounts of field outside the material.



Dividing a ferromagnetic materials into domains reduces the magnetostatic energy.

Size of Domain and Domain Walls

A domain which is too big is unstable, and will divide into smaller domains. The size of domain depends on the balance of several energies within the material. Each time a region of magnetization splits into two domain. It creates a domain wall between domains, it creates a domain wall between domains, where magnetic dipoles with magnetization pointing in different directions are adjacent. Forcing adjacent dipoles to point in different directions require energy. The domains keep dividing into smaller domain until the energy cost of creating an additional domain wall is just equal to field energy saved. Then the domains of this size are stable.

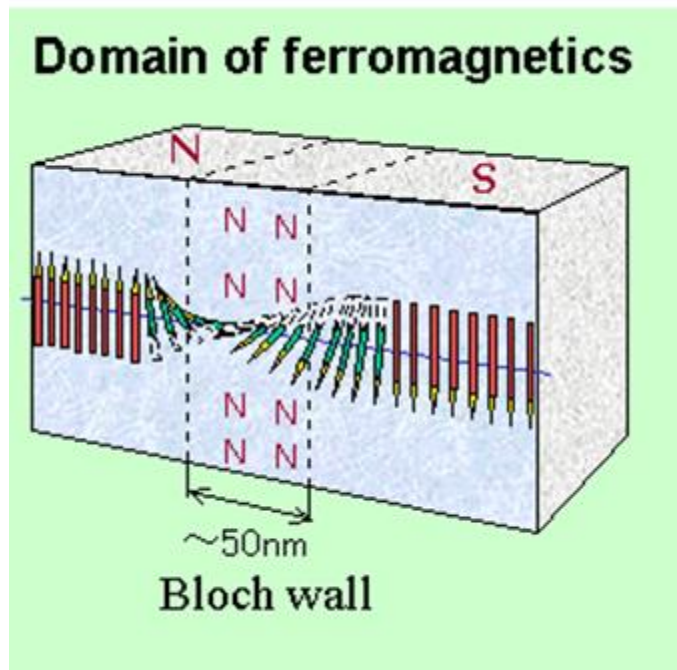


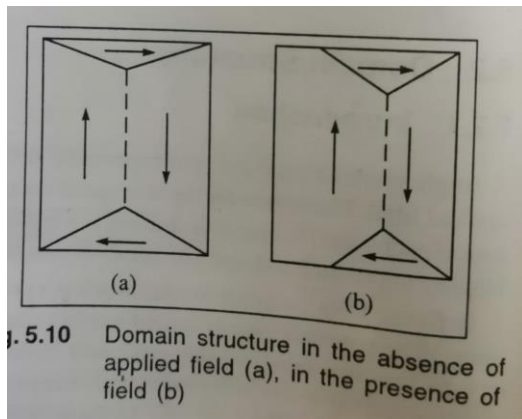
Fig: Domain and domain wall

Exchange forces between neighboring atomic spins favor very little relative motion. Domain wall has neighboring atomic spins rotated gradually and over several hundred atomic spacing the magnetic moments reach a rotation of 180° . If the exchange forces were only responsible then relative rotation would be so little that the wall would have to be so thick. However, magnetic moments that are oriented away from easy directions possess excess energy called anisotropic energy which tries to decrease wall thickness. Wall thickness is a compromise between exchange energy requiring thick wall and anisotropy energy requiring thin wall.

Magnetization of Ferromagnetic materials with external field

Magnetization of the domains is normally along the one of the preferred directions in which the atomic spin alignment is easiest. The magnetization of the crystal along an applied field occurs by growth of the domains with magnetization along the applied field.

The Bloch wall between domains A and B migrates towards right enlarging domain A and shrinking domain B. The migration of Bloch wall is caused by spins in the wall. Also spins in the region B adjacent to the wall being gradually rotated by the field. The magnetization process involves the motion of Bloch walls in the crystal.



B-H Curve and Hysteresis Loop

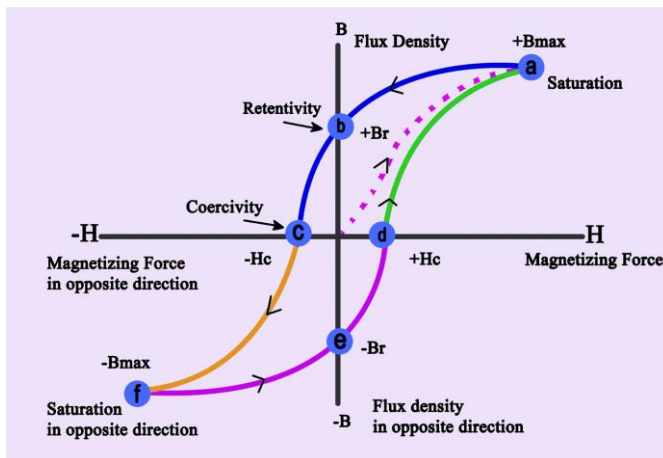


Fig: Hysteresis Loop

In unmagnetized polycrystalline sample, each crystal grain possess domain which depends on shape and size of the grain and magnetization of neighboring grain. When a small field is applied in x-direction, then the domain wall move by small distance and net magnetization is seen on positive x-direction. First there is small net magnetization represented by O_a . As the field is increased, the domain wall motion extends to a larger distance and wall encounters various obstacles such as impurities. A domain wall that is stuck at an imperfection at a given field can't move until field is increased sufficiently to provide necessary force to snap the wall. This sudden jerk in wall motion leads to small jump in magnetization of the specimen as the magnetization field is increased. This phenomenon is called **Barkhausen effect**.

As we continue to increase the field, magnetization increases by jerk domain wall motion and get saturated. If we decrease and remove the magnetization, each grain will rotate to align parallel with the nearest easy direction. There is certain magnetization left after removal of magnetic field called residual magnetization (M_r). If new field is applied in reverse direction magnetization will decrease and becomes zero for certain field (H_c).

Further increase in field in opposite direction cause the effect same as above but in opposite direction.

Eddy Current Loss

Eddy currents are loops of electric current induced within conductors by changing magnetic field in the conductor, due to the Faraday's laws of induction. Eddy current flow in closed loops within conductor, in planes perpendicular to magnetic field.

By Len's law, an eddy current creates a magnetic field that opposes the magnetic field that created it. Eddy currents generate resistive losses that transform some forms of energy, such as kinetic energy into heat. This heating reduces efficiency the iron-core transformers and electric motors and other devices that use changing magnetic fields. Eddy currents are minimized in these devices by selecting magnetic core materials that have low electrical conductivity(eg. Ferrites) or by using thin sheets of magnetic material known as laminations.

Hard And Soft Magnetic Materials

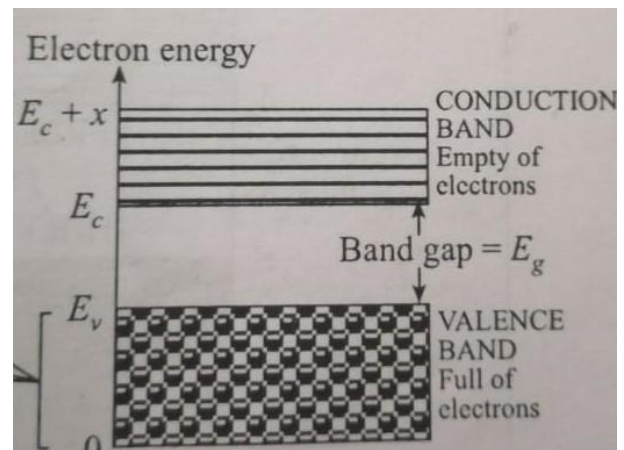
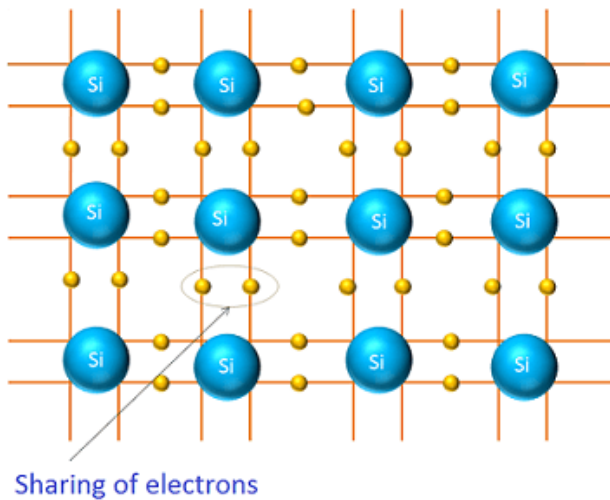
S.N o.	Hard Magnetic Materials	Soft Magnetic Materials
1	Materials which retain their magnetism and are difficult to demagnetize are called hard magnetic materials. These materials retain their magnetism even after the removal of the applied magnetic field. Hence these materials are used for making permanent magnets. In permanent magnets the movement of the domain wall is prevented. They are prepared by heating the magnetic materials to the required temperature and then quenching them. Impurities increase the strength of hard magnetic materials.	Soft magnetic materials are easy to magnetize and demagnetize. These materials are used for making temporary magnets. The domain wall movement is easy. Hence they are easy to magnetize. By annealing the cold worked material, the dislocation density is reduced and the domain wall movement is made easier. Soft magnetic materials should not possess any void and its structure should be homogeneous so that the materials are not affected by impurities.
2	They have large hysteresis loss due to large hysteresis loop area.	They have low hysteresis loss due to small hysteresis area.
3	Susceptibility and permeability are low.	Susceptibility and permeability are high.
4	Coercivity and retentivity values are large.	Coercivity and retentivity values are less.
5	Magnetic energy stored is high.	Since they have low retentivity and coercivity, they are not used for making permanent magnets.
6	They possess high value of BH product.	Magnetic energy stored is less.
7	The eddy current loss is high.	The eddy current loss is less because of high resistivity.

Chapter 6

Semiconducting Materials

Band Structure in Semiconductor

Because of band overlapping, the valance as well as conduction bands of semiconductors consists of mixed s and p-states. Eight highest s+p states of split into two separate(s+p) bands, each of which consists of one s and three p-states. The lower s-stat can accommodate one electron per atom whereas the three lower p states can accommodate 3 electrons per atom. Because germanium and silicon possess four valance electrons per atom, the valance band is completely filled with electrons while conduction band remain empty.



The width of conduction band is denoted by χ and called electron affinity.

Electron and Hole

For a pure silicon crystal covalent bond is saturated and there are no empty states available at valance band at absolute zero temperature. The conduction band is empty, electron from valance band cannot jump to conduction band unless it gets sufficient energy.

When a photon of energy greater than E_g is incident on electron, it moves to conduction band and space left behind in the valance band is called hole. Hence, conduction electron and hole are generated simultaneously.

Also, as the temperature increases, thermal energy causes lattice vibration and bond may deform, an electron hole pair get generated. Due to tunneling, neighboring electron at valance band may occupy the hole and the hole gets shifted in the direction opposite to the direction of electron in conduction band. When electron in conduction band meets the hole of valance band, the electron will occupy empty state releasing some energy and this process is called recombination.

Conduction in Semiconductor

In order for semiconductor to become conducting, electrons have to be excited from valence band into conduction band where they are accelerated by external electric field. Likewise electron holes left behind in valence band migrate in opposite direction to the electrons.

$$\text{Current density}(J) = env_{de} + epv_{dh}$$

Where,

$n = \text{electron concentration}$

$p = \text{hole concentration}$

$v_{de} = \text{drift velocity of electron}$

$v_{dh} = \text{drift velocity of hole}$

Also,

$$v_{de} = \mu_e E_x$$

$$v_{dh} = \mu_h E_x$$

Where, μ_e and μ_h are electron and hole mobility.

$$\text{Hence, } J = en\mu_e E_x + ep\mu_h E_x$$

$$J/E_x = en\mu_e + ep\mu_h$$

$$\sigma = en\mu_e + ep\mu_h$$

Where, σ is electrical conductivity.

Electron and Hole Concentration

Let n be the number of conduction electron per unit volume, $Z(E)$ be the density of states per unit volume and $F(E)$ be the probability of occupation then numbers of electrons in the energy interval E and $E+dE$ is

$$dn = Z(E).F(E)dE$$

$$n = \int_{E_c}^{E_c+\chi} Z(E).F(E)dE$$

Concentration of electron above $E_c+\chi$ is zero, so upper limit can be considered upto ∞ .

$$n = 2 \int_{E_c}^{\infty} Z(E).F(E)dE$$

Also,

$$F(E) = \frac{1}{1 + e^{\frac{E-E_F}{kT}}}$$

$$F(E) \approx e^{\frac{-E+E_F}{kT}}$$

And,

$$Z(E) = C(E - E_c)^{\frac{1}{2}}$$

$$Z(E) = \frac{\pi}{4} k^{\frac{3}{2}} (E - E_c)^{\frac{1}{2}}$$

$$Z(E) = \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} (E - E_c)^{\frac{1}{2}}$$

and concentration of electrons from equation (i) is

$$n = 2 \int_{E_c}^{\infty} \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} (E - E_c)^{\frac{1}{2}} e^{\frac{-E+E_F}{kT}} dE$$

$$n = 2 \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} \int_{E_c}^{\infty} (E - E_c)^{\frac{1}{2}} e^{\frac{-E+E_F}{kT}} dE$$

$$n = 2 \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} \int_{E_c}^{\infty} (E - E_c)^{\frac{1}{2}} e^{\frac{-E-E_c+E_c+E_F}{kT}} dE$$

$$n = 2 \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} \int_{E_c}^{\infty} (E - E_c)^{\frac{1}{2}} e^{\left(\frac{-E-E_c}{kT} \right)} e^{\left(\frac{E_c-E_F}{kT} \right)} dE$$

$$n = 2 \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} e^{\left(\frac{-E_c-E_F}{kT} \right)} \int_{E_c}^{\infty} (E - E_c)^{\frac{1}{2}} e^{\left(\frac{-E-E_c}{kT} \right)} dE$$

Let,

$$\frac{E-E_c}{kT} = x$$

Differentiating both sides with respect to x,

$$dE = kT dx$$

When,

$$E \rightarrow E_c, x \rightarrow 0$$

$$E \rightarrow \infty, x \rightarrow \infty$$

$$n = 2 \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} e^{\left(\frac{-E_c-E_F}{kT} \right)} \int_0^{\infty} (kTx)^{\frac{1}{2}} e^{(-x)} kT dx$$

$$n = 2 \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} e^{\left(\frac{-E_c-E_F}{kT} \right)} (kT)^{\frac{3}{2}} \int_0^{\infty} (x)^{\frac{1}{2}} e^{(-x)} dx$$

We know,

$$\int_0^{\infty} (x)^{\frac{1}{2}} e^{(-x)} dx = \frac{\sqrt{\pi}}{2}$$

$$n = 2 \frac{\pi}{4} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} e^{\left(-\frac{E_c - E_F}{kT} \right)} (kT)^{\frac{3}{2}} \frac{\sqrt{\pi}}{2}$$

$$n = 2 \frac{\pi^2}{8} \left(\frac{8m_e^*}{h^2} \right)^{\frac{3}{2}} e^{\left(-\frac{E_c - E_F}{kT} \right)} (kT)^{\frac{3}{2}}$$

$$n = 2 \left(\frac{2\pi kT m_e^*}{h^2} \right)^{\frac{3}{2}} e^{\left(-\frac{E_c - E_F}{kT} \right)}$$

$$n = N_c e^{\left(-\frac{E_c - E_F}{kT} \right)}$$

Where,

$$N_c = 2 \left(\frac{2\pi kT m_e^*}{h^2} \right)^{\frac{3}{2}} \text{ is effective density of states at conduction band edge.}$$

Subtracting, the probability of occupation function from unity.

$$p = \int_0^{E_v} Z_{VB}(E) (1 - F(E)) dE$$

$$p \approx N_v e^{\left(-\frac{E_F - E_v}{kT} \right)}$$

$$\text{Where, } N_v = 2 \left(\frac{2\pi kT m_h^*}{h^2} \right)^{\frac{3}{2}}$$

Also, numbers of electron = numbers of holes , i.e n=p

$$np = N_c N_v e^{\left(-\frac{E_c - E_F}{kT} \right)} e^{\left(-\frac{E_F - E_v}{kT} \right)}$$

$$n = p = n_i \text{ (for intrinsic concentration)}$$

Where, n_i called as intrinsic concentration.

$$n_i^2 = N_c N_v e^{\left(-\frac{E_c - E_F}{kT} \right)} e^{\left(-\frac{E_F - E_v}{kT} \right)}$$

$$n_i = (N_c N_v)^{\frac{1}{2}} e^{\left(-\frac{E_g}{2kT} \right)}$$

We have known that,

$$n_i = n = p$$

$$\text{So, } N_v e^{\left(-\frac{E_F - E_v}{kT} \right)} = (N_c N_v)^{\frac{1}{2}} e^{\left(-\frac{E_g}{2kT} \right)}$$

Consider, E_{Fi} be denoted as fermi energy level for intrinsic semiconductor.

$$\text{Or, } e^{\left(-\frac{E_{Fi} - E_v}{kT} \right)} = \frac{1}{N_v} (N_c N_v)^{\frac{1}{2}} e^{\left(-\frac{E_g}{2kT} \right)}$$

Taking natural logarithm on both the sides, we will have

$$-\frac{E_{Fi}-E_v}{kT} = \ln\left(\frac{N_c}{N_v}\right)^{\frac{1}{2}} + \left(\frac{-E_g}{2kT}\right)$$

$$-E_{Fi} + E_v = kT \ln\left(\frac{N_c}{N_v}\right)^{\frac{1}{2}} + \left(\frac{-E_g}{2}\right)$$

$$E_{Fi} = E_v - kT \ln\left(\frac{N_c}{N_v}\right)^{\frac{1}{2}} + \left(\frac{E_g}{2}\right)$$

$$E_{Fi} = E_v - kT \ln\left(\frac{\frac{2\pi kT m_e^*}{h^2}}{\frac{2\pi kT m_h^*}{h^2}}\right)^{\frac{3}{4}} + \left(\frac{E_g}{2}\right)$$

$$E_{Fi} = E_v - \frac{3}{4} kT \ln\left(\frac{m_e^*}{m_h^*}\right) + \left(\frac{E_g}{2}\right)$$

Thus, if effective masses of electron in conduction band and of hole in valance band are equal to each other, the intrinsic fermi level in semiconductor lies in the middle of energy gap.

$$E_{Fi} = E_v + \frac{E_g}{2}$$

$$\text{Also, } E_{Fi} = E_c - \frac{E_g}{2}$$

Normally, effective masses of conduction electron and valence hole differ slightly, which means the intrinsic fermi level does not long precisely at the center of the energy band gap, but is shifted a little bit. In Si the hole effective mass is slightly greater than that of the electron. So, the intrinsic fermi level is slightly above the mid-gap.

Extrinsic Semiconductors

In order to modulate the electrical properties of semiconductor atoms of foreign elements belonging to III A and V A group are introduced intentionally in pure silicon crystal. This process is called doping and resulting semiconductor is called extrinsic semiconductor.

n-type semiconductor

On adding pentavalent impurities into silicon such as arsenic, it is forced to bond with neighboring silicon atom as in diamond crystal structure. Only four electrons will participate in bonding and 5th electron continuously goes on orbiting to arsenic (As⁺) ion core similar to electron in hydrogen atom. Hence, ionization energy required to free the electron is,

$$\text{For hydrogen, } E_b = E_1 = \frac{e^4 m_e}{8\epsilon_0^2 h^2} = -13.6 \text{ eV}$$

Since, the electron is orbiting in the silicon environment hence we have to use absolute permittivity of silicon and effective mass of electron in silicon,

$$E_b^{Si} = \frac{e^4 m_e^*}{8\epsilon_0^2 \epsilon_r^2 h^2} = -0.032 \text{ eV}$$

Where, $m_e^* = \frac{m_e}{3}$ and $\epsilon_r = 11.9$

Also, average thermal energy of atomic vibration is $3kT$ and nearly equal to 0.07eV .

Thus, this 5th electron normally found in conduction band.

In n-type semiconductor, the conduction electrons are majority charge carriers and holes are minority charge carriers. If n and p are electron and hole concentration and N_d is donor concentration, then

$$n = N_d + p$$

Also, we have

$$n_i^2 = np$$

$$p = \frac{n_i^2}{n}$$

$$n = N_d + \frac{n_i^2}{n}$$

$$n^2 - nN_d - n_i^2 = 0$$

$$n = \frac{N_d}{2} \pm \sqrt{\left(\frac{N_d^2}{4} + n_i^2\right)}$$

But, $n_i^2 \ll \frac{N_d^2}{4}$ then

$$n = \frac{N_d}{2} + \frac{N_d}{2} \text{ (Negative value is discarded because electron concentration cannot be negative)}$$

$$n = N_d$$

$$p = \frac{n_i^2}{N_d}$$

Hole concentration in n-type semiconductor is less than intrinsic hole concentration because some of the electron in conduction band recombine with hole to maintain $n_i^2 = np$. This is called minority carrier suppression.

$$\sigma = en\mu_e + ep\mu_h$$

$$\sigma = eN_d\mu_e + e\frac{n_i^2}{N_d}\mu_h$$

$$\sigma \approx eN_d\mu_e$$

So, conductivity is mainly due to mobility of electrons in n-type semiconductor.

p-type Semiconductor

On adding trivalent impurities into silicon such as boron, it is forced to bond with neighboring silicon atom. Only three electrons will participate in bonding and 4th is not present. That is why, hole formation takes place in the valence band.

The binding energy is very small (0.05eV). So at room temperature all the acceptor sites are ionized by thermal vibrations in the lattice. When the boron accepts electron from neighboring Si-Si bond, the hole is shifted away from the boron to Si-Si bond in valence bond. The boron atom accepts an electron from Si-Si bond, and resulting in semiconductor called p-type semiconductor.

If n and p be the electron and hole concentration in the semiconductor and N_a be the acceptor concentration.

Then, hole concentration is given by,

$$p = N_a + n$$

$$n_i^2 = np$$

$$n = \frac{n_i^2}{p}$$

$$p = N_a + \frac{n_i^2}{p}$$

$$p^2 - pN_a - n_i^2 = 0$$

$$p = \frac{N_a}{2} \pm \sqrt{\left(\frac{N_a^2}{4} + n_i^2\right)}$$

$$\text{Since, } n_i^2 \ll \frac{N_a^2}{4}$$

$$\text{Thus, } p \approx \frac{N_a}{2} + \frac{N_a}{2} = N_a$$

The conductivity in p-type semiconductor is mainly due to hole concentration which can be calculated,

$$\sigma = en\mu_e + ep\mu_h$$

$$\sigma = eN_a\mu_h + e\frac{n_i^2}{N_a}\mu_e$$

$$\sigma \approx eN_a\mu_h$$

Compensation Doping

Adding trivalent impurities will increase hole concentration in the semiconductor whereas by adding pentavalent impurities to the same will compensate for the increase in hole concentration and decrease in electron concentration. Assuming all donor impurities and acceptor impurities ionized at room temperature, the effective hole concentration and electron concentration are calculated based on which of the impurity is greater in concentration.

In N_a and N_d are acceptor and donor concentration respectively. If $N_d > N_a$, then holes due to acceptor doping will recombine with the electrons due to donor doping. So, the electron concentration is,

$$n = N_d - N_a$$

$$p = \frac{n_i^2}{N_d - N_a}$$

Similarly, when $N_a > N_d$, the electrons due to donor doping recombine with holes due to acceptor doping and the resultant hole concentration are,

$$p = N_a - N_d$$

$$n = \frac{n_i^2}{N_a - N_d}$$

Where, n_i is intrinsic concentration of semiconductor.

Diffusion in Semiconductor

Diffusion is the process in which particle tends to spread out and distribute themselves as a result of their random thermal vibration migrating from region of high concentration to region of low concentration.

In a semiconductor, when doping is done then there is more concentration of one type of carrier. So it gets diffused i.e get distributed throughout the semiconductor. Since the particles are charged, diffusion of charge causes rise of electrical current called diffusion current.

$$\text{Diffusion Current density } (J_{e-diff}) = eD_e \frac{dn}{dx} \text{----- (i)}$$

$$\text{And, } (J_{n-diff}) = -eD_h \frac{dp}{dx} \text{----- (ii)}$$

Where, D_e and D_h are electron and hole diffusion coefficients respectively measured in cm^2/sec .

The diffusion currents are directly proportional to the carrier concentration gradients.

For diffusion to occur,

$$\frac{dn}{dx} \neq 0 \text{ and } \frac{dp}{dx} \neq 0 \text{ (for 1-dimension)}$$

For 3D

$$\nabla p \neq 0 \text{ and } \nabla n \neq 0$$

Equation (i) and (ii) is called Fick's first law of diffusion in one dimension.

Total current density in semiconductor

= Total current density due to drift + total current density due to diffusion

$$\text{i.e } J = J_{e-drift} + J_{h-drift} + J_{e-diff} + J_{h-diff}$$

Degenerate and Non-degenerate Semiconductor

We have concentration of electron in conduction band

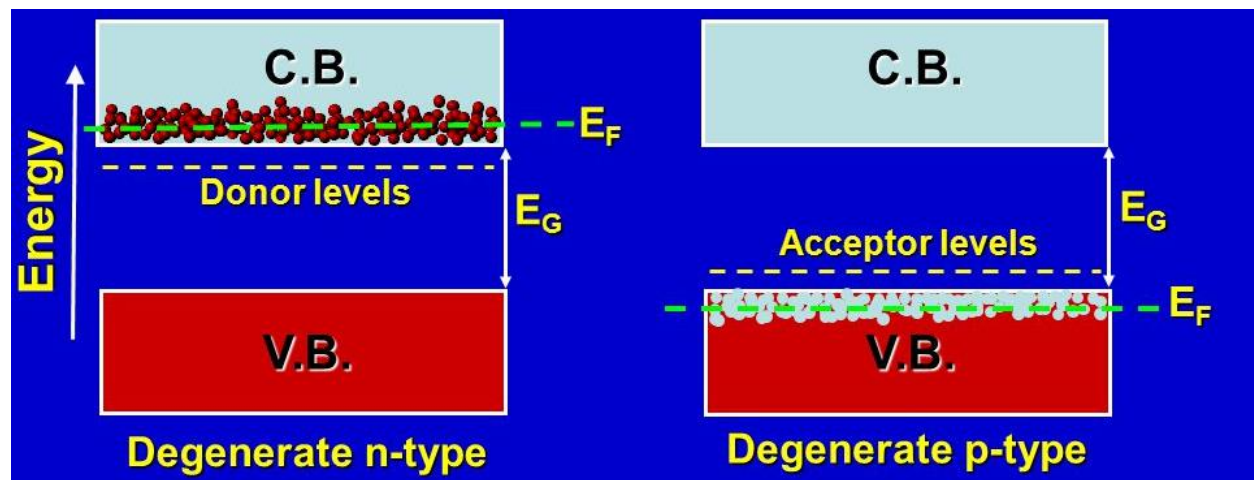
$$n = N_c e^{\left(-\frac{E_c - E_F}{kT}\right)}$$

Which is derived by replacing Fermi-Dirac distribution with Boltzmann's statistics which is valid if $E_c \gg kT$ above E_F .

We assume number of states is far greater than number of electrons. So, we can assume no two electrons are found in the same state. It neglects Pauli's exclusion principle. Boltzmann statistics is valid if number of states is very very greater than numbers of electrons in conduction band. Semiconductor with $n \ll N_c$ and $p \ll N_v$ are termed as non-degenerate semiconductor.

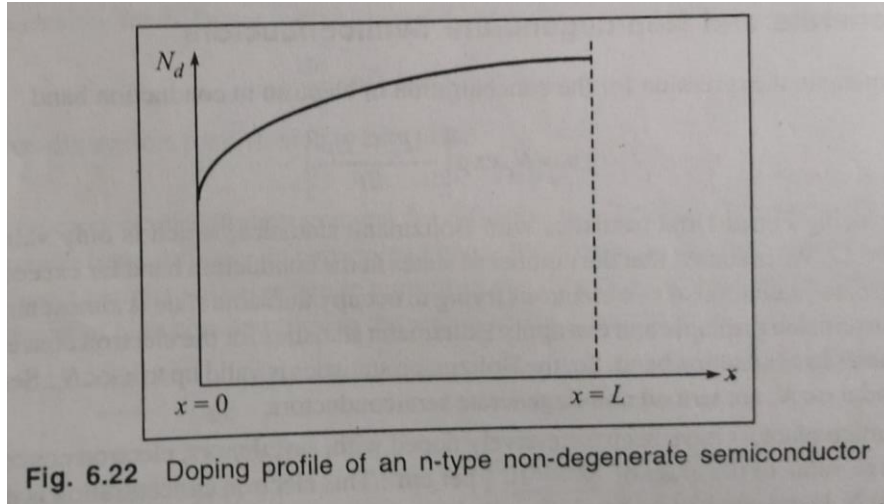
When the semiconductor are highly doped then concentration of electron gets nearly equal to the numbers of states N_c , in conduction band. In such a case Pauli's exclusion principle cannot be neglected and Fermi Dirac distribution has to be used. Such semiconductor shows some properties similar to that of metal. Semiconductor with $n \gg N_c$ and $p \gg N_v$ are termed as degenerate semiconductor.

In n-type semiconductor, when donor orbitals overlap to form narrow energy band, it overlaps and becomes part of conduction band thereby slightly shifting E_c downward. Fermi level in n-type degenerate semiconductor is within the conduction band. In p-type degenerate semiconductor, Fermi energy lies below E_v .

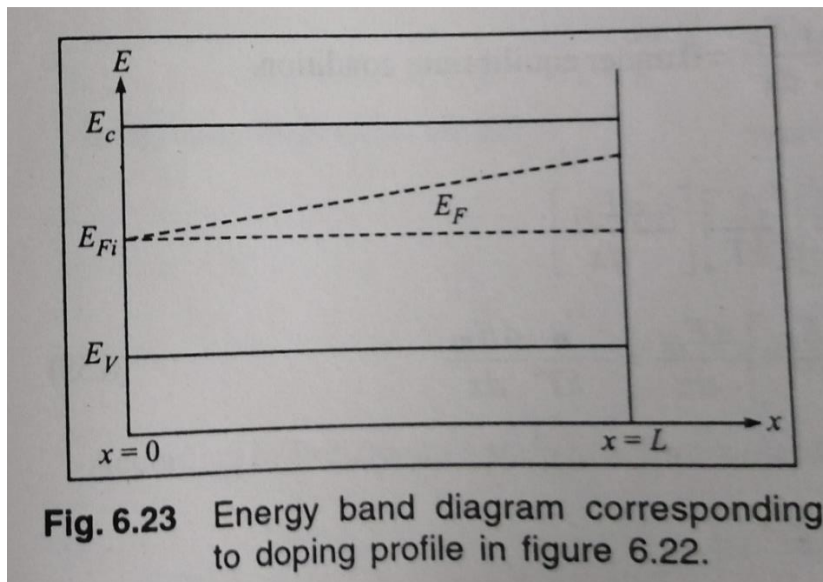


Einstein Relationship

For a non-degenerate semiconductor the doping profile along its length is given by fig below.



So, we expect corresponding energy band diagram to be as shown in figure below.



The above diagram shows the variation in Fermi energy level along length. But in equilibrium Fermi energy level inside a material cannot vary with respect to position.

Mathematically,

$$\frac{dE_F}{dx} = \frac{dE_F}{dy} = \frac{dE_F}{dz} = 0$$

To keep Fermi energy level constant, other energy band must bend which is commonly known as band bending in non-uniformly doped semiconductors. So, the band energy diagram looks like,

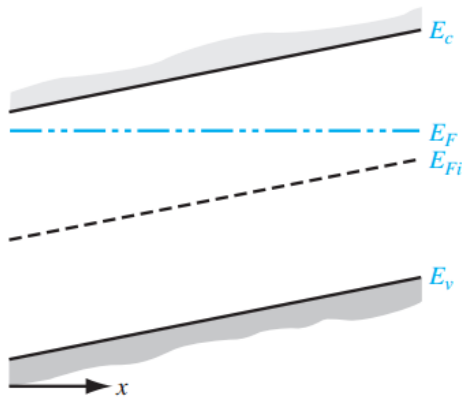


Figure 5.12 | Energy-band diagram for a semiconductor in thermal equilibrium with a nonuniform donor impurity concentration.

Under equilibrium condition, current in semiconductor must be zero.

$$J = J_{e-drift} + J_{e-diff} = 0$$

$$en\mu_e \vec{E} + eD_e \frac{dn}{dx} = 0 \text{ ----- (i)}$$

$$\text{Potential energy } PE = -eV \text{ ----- (ii)}$$

With respect to some reference E_{ref} , potential energy is given as,

$$PE = E_c - E_{ref} \text{ ----- (iii)}$$

From (ii) and (iii)

$$V = -\frac{1}{e}(E_c - E_{ref}) \text{ ----- (iv)}$$

$$\text{Electric field } \vec{E} = -\nabla V$$

$$\text{In one dimension, } \vec{E} = -\frac{dV}{dx}$$

$$\vec{E} = \frac{dV}{dx} \left[\frac{1}{e}(E_c - E_{ref}) \right]$$

$$\vec{E} = \frac{1}{e} \left(\frac{dE_c}{dx} - \frac{dE_{ref}}{dx} \right)$$

$$\vec{E} = \frac{1}{e} \frac{dE_c}{dx} \quad [E_{ref} \text{ is constant}]$$

$$\text{Therefore, } e\vec{E} = \frac{dE_c}{dx} = \frac{dE_v}{dx} = \frac{dE_{Fi}}{dx}$$

To find $\frac{dn}{dx}$, we have

$$n = N_c e^{\left(\frac{E_c - E_F}{kT}\right)} \text{ (for n-type semiconductor)}$$

$$n_i = N_c e^{\left(\frac{E_c - E_{Fi}}{kT}\right)}$$

$$\frac{n}{n_i} = \frac{N_c e^{\left(\frac{E_c - E_F}{kT}\right)}}{N_c e^{\left(\frac{E_c - E_{Fi}}{kT}\right)}}$$

$$\frac{n}{n_i} = e^{\left(\frac{E_c - E_F}{kT}\right)} \times e^{\left(\frac{E_c - E_{Fi}}{kT}\right)}$$

$$n = n_i e^{\left(\frac{-E_c + E_F + E_c - E_{Fi}}{kT}\right)}$$

$$n = n_i e^{\left(\frac{E_F - E_{Fi}}{kT}\right)} \text{ ----- (v)}$$

Under equilibrium condition,

$$\frac{dE_F}{dx} = 0$$

Differentiating equation (v) on both sides with respect to x,

$$\frac{dn}{dx} = n_i \frac{de^{\left(\frac{E_F - E_{Fi}}{kT}\right)}}{dx}$$

$$\frac{dn}{dx} = n_i e^{\left(\frac{E_F - E_{Fi}}{kT}\right)} \times \frac{1}{kT} \left[\frac{dE_F}{dx} - \frac{dE_{Fi}}{dx} \right]$$

$$\frac{dn}{dx} = n_i e^{\left(\frac{E_F - E_{Fi}}{kT}\right)} \times \frac{1}{kT} \left[0 - \frac{dE_{Fi}}{dx} \right]$$

$$\frac{dn}{dx} = \frac{n}{kT} \left[-\frac{dE_{Fi}}{dx} \right] \text{ ----- (vi) [from equation (v)]}$$

We have,

$$\frac{dE_{Fi}}{dx} = e\vec{E} \text{ ----- (vii)}$$

Using equation (vi), (vii) and equation (i) we get,

$$en\mu_e \vec{E} + eD_e \left[-\frac{n}{kT} e\vec{E} \right] = 0$$

$$en \left[\mu_e - \frac{D_e}{kT} e \right] \vec{E} = 0$$

Which implies

$$\mu_e - \frac{D_e}{kT} e = 0$$

$$\mu_e = \frac{D_e}{kT} e$$

$$\mu_e kT = D_e e$$

$$\frac{kT}{e} = \frac{D_e}{\mu_e}$$

This is required Einstein relationship relating electron diffusion coefficient to electron mobility.

At constant temperature ratio of diffusion coefficient to electron mobility is constant and equal to $\frac{kT}{e}$.

Recombination and Generation

A. Direct Recombination and Generation

Above absolute zero temperature, the thermal electrons from the valence band to conduction band continuously generates electron hole pairs. So, when a wandering free electron in conduction band falls down to low energy state and meets the hole by releasing the excess energy(as photon). This process is called recombination/direct recombination.

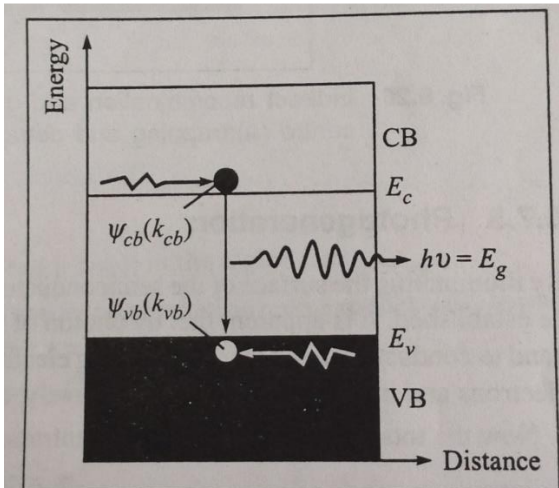
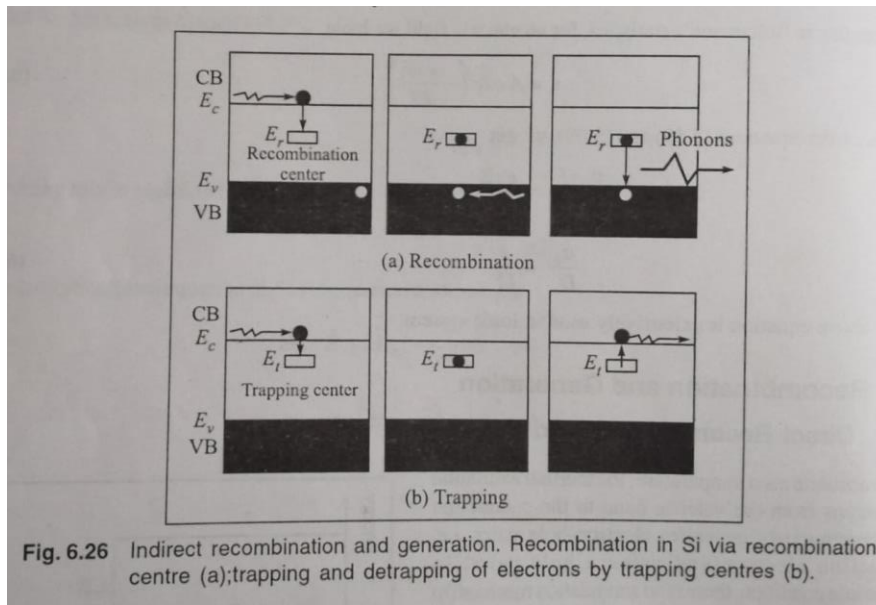


Fig. 6.25 Direct recombination in GaAs

B. Indirect Recombination and Generation

Indirect recombination occurs through the recombination due to the impurity or crystal defect which can take up any momentum difference between a hole and electron. The electron is captured at the recombination center and are trapped until a hole arrives and recombines with it. Recombination center provides a localized electronic state below E_c in the band gap. When an electron(e^-) approaches the center, it is captured and is found. The localized electron waits on the center and when it finds the hole, it recombines in the valence band.



PN Junction

We know, how to make an N-type semiconductor material by doping a silicon atom with small amounts of Antimony and also how to make a P-type semiconductor material by doping another silicon atom with Boron.

This is all well and good, but these newly doped N-type and P-type semiconductor materials do very little on their own as they are electrically neutral. However, if we join (or fuse) these two semiconductor materials together they behave in a very different way merging together and producing what is generally known as a “**PN Junction**”.

When the N-type semiconductor and P-type semiconductor materials are first joined together a very large density gradient exists between both sides of the PN junction. The result is that some of the free electrons from the donor impurity atoms begin to migrate across this newly formed junction to fill up the holes in the P-type material producing negative ions.

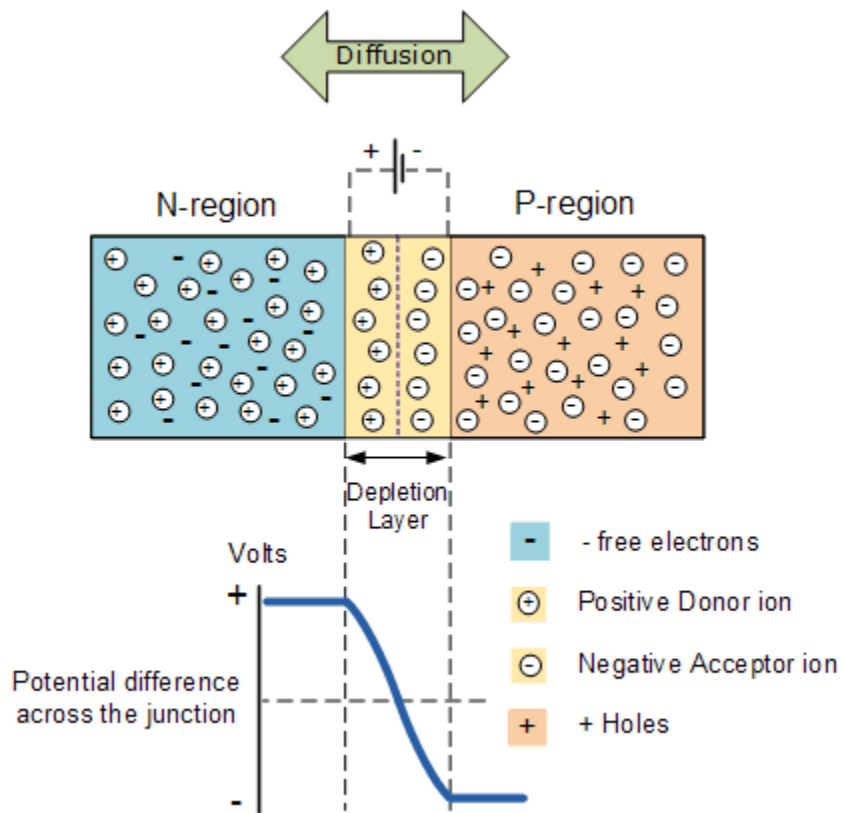
However, because the electrons have moved across the PN junction from the N-type silicon to the P-type silicon, they leave behind positively charged donor ions (N_D) on the negative side and now the holes from the acceptor impurity migrate across the junction in the opposite direction into the region where there are large numbers of free electrons.

As a result, the charge density of the P-type along the junction is filled with negatively charged acceptor ions (N_A), and the charge density of the N-type along the junction becomes positive. This charge transfer of electrons and holes across the PN junction is known as **diffusion**. The width of these P and N layers depends on how heavily each side is doped with acceptor density N_A , and donor density N_D , respectively.

This process continues back and forth until the number of electrons which have crossed the junction have a large enough electrical charge to repel or prevent any more charge carriers from crossing over the junction. Eventually a state of equilibrium (electrically neutral situation) will occur producing a

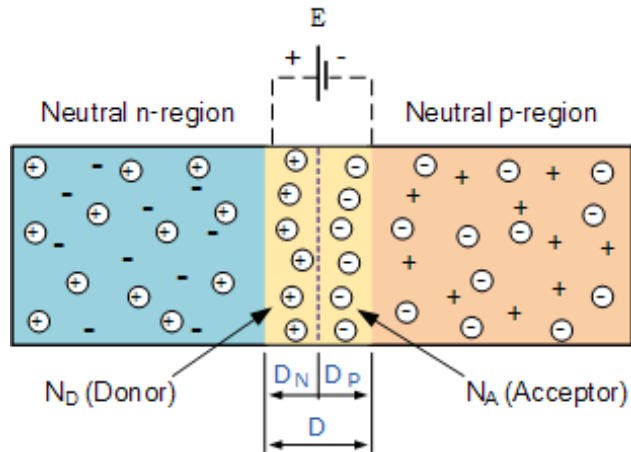
“potential barrier” zone around the area of the junction as the donor atoms repel the holes and the acceptor atoms repel the electrons.

Since no free charge carriers can rest in a position where there is a potential barrier, the regions on either sides of the junction now become completely depleted of any more free carriers in comparison to the N and P type materials further away from the junction. This area around the **PN Junction** is now called the **Depletion Layer**.



Depletion Region Details

- In the p-type region, there are holes from acceptor impurities and in the n-type region there are extra electrons.
- When a PN junction is formed, some of the electrons from n-type region which have reached the conduction band are free to diffuse across the junction and combine with holes.
- Filling a hole makes a negative ion and leaves behind a positive ion on N-side. A space charge builds up, creating a depletion region.



Let us consider,

$D_p \rightarrow$ width of depletion region in p – type

$D_n \rightarrow$ width of depletion region in n – type

$D \rightarrow$ total width of depletion region

$N_a \rightarrow$ acceptor concentration in p – region

$N_d \rightarrow$ acceptor concentration in n – region

$n_{n_0} \rightarrow$ electron concentration in n – region

$p_{n_0} \rightarrow$ hole concentration in n – region

$n_{p_0} \rightarrow$ electron concentration in p – region

$p_{p_0} \rightarrow$ hole concentration in p – region

Also, we know

$$n_{n_0} = N_d$$

$$p_{n_0} = \frac{n_i^2}{n_{n_0}} = \frac{n_i^2}{N_d}$$

$$p_{p_0} = N_a$$

$$n_{p_0} = \frac{n_i^2}{p_{p_0}} = \frac{n_i^2}{N_a}$$

The exposed charge in the left side of the junction must be equal to exposed charge in RHS to maintain charge neutrality.

$$\frac{N_a}{N_d} = \frac{D_n}{D_p}$$

$$D_p N_a = D_n N_d$$

We have,

$$E = -\frac{eN_d D_n}{\epsilon} = -\frac{eN_a D_p}{\epsilon} \text{----- (i)}$$

We know that, potential at any point can be calculated from electric field as,

$$E = -\frac{dv}{dx}$$

We know that sum of net current due to drift and diffusion is zero.

$$J_{e-drift} + J_{e-diff} = 0$$

$$en\mu_e E + eD_e \frac{dn}{dx} = 0$$

$$en\mu_e E = -eD_e \frac{dn}{dx}$$

$$E = -\frac{D_e}{\mu_e} \frac{dn}{dx}$$

From Einstein Relationship

$$\frac{D_e}{\mu_e} = \frac{kT}{e}$$

Substituting $\frac{D_e}{\mu_e}$ in above equation we get,

$$E = -\frac{kT}{e} \frac{dn}{dx}$$

Also,

$$E = -\frac{dv}{dx}$$

Thus,

$$-\frac{dv}{dx} = -\frac{kT}{e} \frac{dn}{dx}$$

$$dv = \frac{kT}{e} \frac{dn}{n}$$

Integrating from $-D_p$ to D_n ,

$$\int_{-D_p}^{D_n} dv = \frac{kT}{e} \int_{-D_p}^{D_n} \frac{dn}{n}$$

$$V(D_n) - V(-D_p) = \frac{kT}{e} [\ln(D_n) - \ln(-D_p)]$$

$$V_0 = \frac{kT}{e} [\ln(D_n) - \ln(-D_p)]$$

Where,

$n(D_n)$ is electron concentration at (D_n) .

At, D_n , $n=N_d$

At $-D_p$, $n = \frac{n_i^2}{N_a}$

$$V_0 = \frac{kT}{e} [\ln(N_d) - \ln\left(\frac{n_i^2}{N_a}\right)]$$

$$V_0 = \frac{kT}{e} \ln\left[\frac{N_d \cdot N_a}{n_i^2}\right]$$

This is the expression for the built in potential in PN junction.

$$V_0 = \frac{kT}{e} \ln\left[\frac{n_i^2}{n_{p0}} \times \frac{n_{n0}}{n_i^2}\right]$$

$$V_0 = \frac{kT}{e} \ln\left[\frac{n_{n0}}{n_{p0}}\right]$$

Similarly,

$$V_0 = \frac{kT}{e} \ln\left[\frac{p_{p0}}{p_{n0}}\right]$$

For the width of PN junction

In the above expression,

$n_{n0} \rightarrow$ electron concentration in n – region

$p_{n0} \rightarrow$ hole concentration in n – region

$n_{p0} \rightarrow$ electron concentration in p – region

$p_{p0} \rightarrow$ hole concentration in p – region

Also, the built in potential = $-\frac{eD}{2}$

Where, $D = D_n + D_p$ and $D_p = D - D_n$

Also, we have

$$D_n N_d = D_p N_a$$

$$D_n N_d = (D - D_n) N_a$$

$$D_n N_d + D_n N_a = D N_a$$

$$D_n = \frac{D N_a}{N_d + N_a}$$

Similarly,

$$D_p = \frac{D N_d}{N_d + N_a}$$

Substituting value of D_p in equation (i)

$$E = -\frac{eN_aDN_d}{\epsilon(N_a+N_d)}$$

Now, we have

$$V_0 = -\frac{ED}{2}$$

$$V_0 = \frac{eN_aD^2N_d}{2\epsilon(N_a+N_d)}$$

$$D^2 = \frac{2\epsilon(N_a+N_d)V_0}{eN_aN_d}$$

$$D = \sqrt{\frac{2\epsilon(N_a+N_d)V_0}{eN_aN_d}}$$

$$D = \sqrt{\frac{2\epsilon V_0}{e} \left(\frac{1}{N_a} + \frac{1}{N_d} \right)}$$

Above is the required expression for the width of the depletion layer.

(Write about forward and reverse biasing of PN junction yourself.)

Chapter 7

Semiconductor Material Procession

Crystal Growing

The first step in integrated circuit (IC) fabrication is preparing the high purity single crystal Si wafer. This is the starting input to the fab. Typically, Si wafer refers to a single crystal of Si with a specific orientation, dopant type, and resistivity (determined by dopant concentration). Typically, Si (100) or Si (111) wafers are used. The numbers (100) and (111) refers to the orientation of the plane parallel to the surface. The wafer should have structural defects, like dislocations, below a certain permissible level and impurity (undesired) concentration of the order of ppb (parts per billion).

There are two main techniques for converting polycrystalline EGS (Electronic Grade Si) into a single crystal ingot, which are used to obtain the final wafers.

1. Czochralski technique (CZ)
This is the dominant technique for manufacturing single crystals. It is especially suited for the large wafers that are currently used in IC fabrication.
2. Float zone technique
This is mainly used for small sized wafers. The float zone technique is used for producing specialty wafers that have low oxygen impurity concentration.

Czochralski crystal growth technique

The various components of the process are

1. Furnace
2. Crystal pulling mechanism
3. Ambient control - atmosphere
4. Control system

The starting material for the CZ process is electronic grade silicon, which is melted in the furnace. To minimize contamination, the crucible is made of SiO_2 or SiN_x . The drawback is that at the high temperature the inner liner of the crucible also starts melting and has to be replaced periodically. The furnace is heated above 1500°C , since Si melting point is 1412°C . A small seed crystal, with the desired orientation of the final wafer, is dipped in the molten Si and slowly withdrawn by the crystal pulling mechanism. The seed crystal is also rotated while it is being pulled, to ensure uniformity across the surface. The furnace is rotated in the direction opposite to the crystal puller. The molten Si sticks to the seed crystal and starts to solidify with the same orientation as the seed crystal is withdrawn. Thus, a single crystal ingot is obtained. To create doped crystals, the dopant material is added to the Si melt so that it can be incorporated in the growing crystal. The process control, i.e. speed of withdrawal and the speed of rotation of the crystal puller, is crucial to obtain a good quality single crystal. There is a feedback system that control this process. Similarly there is another ambient gas control system. The final solidified Si obtained is the single crystal ingot. A 450 mm wafer ingot can be as heavy as 800 kg.

Float zone technique

The float zone technique is suited for small wafer production, with low oxygen impurity. A polycrystalline EGS rod is fused with the single crystal seed of desired orientation. This is taken in an inert gas furnace and then melted along the length of the rod by a traveling radio frequency (RF) coil. The RF coil starts from the fused region, containing the seed, and travels up. When the molten region solidifies, it has the same orientation as the seed. The furnace is filled with an inert gas like argon to reduce gaseous impurities. Also, since no crucible is needed it can be used to produce oxygen 'free' Si wafers. The difficulty is to extend this technique for large wafers, since the process produces large number of dislocations. It is used for small specialty applications requiring low oxygen content wafers.

Doping by Solid State Diffusion

- Diffusion is the movement of impurity atoms in a semiconductor material at high temperatures.
- The driving force of diffusion is the concentration gradient.
- There is a wide range of diffusivities for the various dopant species, which depend on how easy the respective dopant impurity can move through the material.
- Diffusion is applied to anneal the crystal defects after ion implantation or to introduce dopant atoms into silicon from a chemical vapor source.
- In the last case, the diffusion time and temperature determine the depth of dopant penetration.
- Diffusion is used to form the source, drain, and channel regions in an MOS transistor.
- But diffusion can also be an unwanted parasitic effect because it takes place during all high-temperature (900 – 1200°C) process steps.

The mathematical expression of diffusion process in 1-dimension can be derived using Fick's law of diffusion. Fick's first law of diffusion states that the particle flow per unit area (particle flux), J , is directly proportional to concentration gradient of the particle.

$$J = -D \frac{\partial N}{\partial x}$$

Where, N is the particle concentration and D is the diffusion coefficient. The –ve sign indicates that particles move from region of high concentration to low concentration.

The continuity equation for particle flux is $\frac{\partial N}{\partial x} = -\frac{\partial J}{\partial x}$

Combining above equations gives Fick's 2nd law of diffusion.

$$\frac{\partial N}{\partial x} = D \frac{\partial^2 N}{\partial x^2}$$

Ion Implantation

- Ion Implantation is the process of adding impurities to a silicon wafer.
- This is performed with an electric field which accelerates the ionized atoms or molecules so that these particles penetrate into the target material until they come to rest because of interactions with the silicon atoms.
- Ion implantation is able to control exactly the distribution and dose of the dopants in silicon because the penetration depth depends on the kinetic energy of the ions which is proportional to the electric field. The dopant dose can be controlled by varying the ion source.
- Unfortunately, after ion implantation, the crystal structure is damaged this implies worse electrical properties.
- Another problem is that the implanted dopants are electrically inactive because they are situated on interstitial sites.
- Therefore after ion implantation, a thermal process step is necessary which repairs the crystal damage and activates the dopants.

Photolithography

- Lithography is used to transfer a pattern from a photomask to the surface of the wafer.
- Photolithography is the process of creating patterns on a smooth surface (Silicon wafer).
- This is accomplished by selectively exposing parts of the wafer while other parts are protected. The exposed sections are susceptible to doping, removal, or metallization. Specific patterns can be created to form regions of conductors, insulators, or doping. Putting these patterns onto a wafer is called photolithography.
- The pattern defined by the mask is either removed or remained after development, depending if the type of resist is positive or negative.

Planar Process

Step 1

Wafer production

The first step is wafer production. The wafer is a round slice of semiconductor material such as silicon. Silicon is preferred due to its characteristics. It is more suitable for manufacturing IC. It is the base or substrate for entire chip. First purified polycrystalline silicon is created from the sand. Then it is heated to produce molten liquid. A small piece of solid silicon is dipped on the molten liquid. Then the solid silicon (seed) is slowly pulled from the melt. The liquid cools to form single crystal ingot. A thin round wafer of silicon is cut using wafer slicer. Wafer slicer is a precise cutting machine and each slice having thickness about .01 to .025 inches. When wafer is sliced, the surface will be damaged. It can be smoothening by polishing. After polishing the wafer, it must thoroughly clean and dried. The wafers are cleaned using high purity low particle chemicals. The silicon wafers are exposed to ultra pure oxygen.

Epitaxial growth

It means the growing of single silicon crystal upon original silicon substrate. A uniform layer of silicon dioxide is formed on the surface of wafer.

Step 2

Masking

To protect some area of wafer when working on another area, a process called **photolithography** is used. The process of photolithography includes masking with a photographic mask and photo etching. A photoresist film is applied on the wafer. The wafer is aligned to a mask using photo aligner. Then it is exposed to ultraviolet light through mask. Before that the wafer must be aligned with the mask. Generally, there are automatic tools for alignment purpose.

Step 3

Etching

It removes material selectively from the surface of wafer to create patterns. The pattern is defined by etching mask. The parts of material are protected by this etching mask. Either wet (chemical) or dry (physical) etching can be used to remove the unmasked material. To perform **etching** in all directions at same time, isotropic etching will be used. Anisotropic etching is faster in one direction. Wet etching is isotropic, but the etching time control is difficult. Wet etching uses liquid solvents for removing materials. It is not suited to transfer pattern with submicron feature size. It does not damage the material. Dry etching uses gases to remove materials. It is strongly anisotropic. But it is less selective. It is suited to transfer pattern having small size. The remaining photo resist is finally removed using additional chemicals or plasma. Then the wafer is inspected to make sure that the image is transferred from mask to the top layer of wafer.

Step 4

Doping

To alter the electrical character of silicon, atom with one less electron than silicon such as boron and atom with one electron greater than silicon such as phosphorous are introduced into the area. The P-type (boron) and N-type (phosphorous) are created to reflect their conducting characteristics. Diffusion is defined as the movement of impurity atoms in semiconductor material at high temperature.

Atomic diffusion

In this method p and n regions are created by adding dopants into the wafer. The wafers are placed in an oven which is made up of quartz and it is surrounded with heating elements. Then the wafers are heated at a temperature of about 1500-2200°F. The inert gas carries the dopant chemical. The dopant and gas is passed through the wafers and finally the dopant will get deposited on the wafer. This method can only be used for large areas. For small areas it will be difficult and it may not be accurate.

Ion implantation

This is also a method used for adding dopants. In this method, dopant gas such as phosphine or boron trichloride will be ionized first. Then it provides a beam of high energy dopant ions to the specified regions of wafer. It will penetrate the wafer. The depth of the penetration depends on the energy of the beam. By altering the beam energy, it is possible to control the depth of penetration of dopants into the wafer. The beam current and time of exposure is used to control the amount of dopant. This method is slower than atomic diffusion process. It does not require masking and this process is very precise. First it points the wafer that where it is needed and shoot the dopants to the place where it is required.

Step 5

Metallization

It is used to create contact with silicon and to make interconnections on chip. A thin layer of aluminum is deposited over the whole wafer. Aluminum is selected because it is a good conductor, has good mechanical bond with silicon, forms low resistance contact and it can be applied and patterned with single deposition and etching process.

Making successive layers: - The process such as masking, etching, doping will be repeated for each successive layers until all integrated chips are completed. Between the components, silicon dioxide is used as insulator. This process is called chemical vapor deposition. To make contact pads, aluminum is deposited. The fabrication includes more than three layers separated by dielectric layers. For electrical and physical isolation a layer of solid dielectric is surrounded in each component which provides isolation. It is possible to fabricate PNP and NPN transistor in the same silicon substrate. To avoid damage and contamination of circuit, final dielectric layer (passivation) is deposited. After that, the individual IC will be tested for electrical function. Check the functionality of each chip on wafer. Those chips are not passed in the test will be rejected.

Assembly and packaging

Each of the wafers contains hundreds of chips. These chips are separated and packaged by a method called scribing and cleaving. The wafer is similar to a piece of glass. A diamond saw cut the wafer into single chips. The diamond tipped tool is used to cut the lines through the rectangular grid which separates the individual chips. The chips that are failed in electrical test are discarded. Before packaging, remaining chips are observed under microscope. The good chip is then mounted into a package. Thin wire is connected using ultrasonic bonding. It is then encapsulated for protection. Before delivered to customer, the chip is tested again. There are three configurations available for packaging. They are metal can package, ceramic flat package and dual in line package. For military applications, the chip is assembled in ceramic packages. The complete integrated circuits are sealed in anti-static plastic bags.

